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PTQNet: Periodic-temporal query network for long-term multivariate time series forecasting ☆

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ABSTRACT

Long-term multivariate time series forecasting remains challenging due to the entanglement of periodic structures and irregular dynamics, especially under high-dimensional settings. This paper proposes **P**eriodic-**T**emporal **Q**uery **Net**work(PTQNet), a lightweight architecture that explicitly decomposes multi-scale seasonal patterns from residual variations. Two core contributions of PTQNet involve: a hierarchical cycle module that learns data-driven periodic embeddings to capture structured seasonality over a generic bank of candidate cycle lengths and a phase-aware residual attention mechanism that dynamically modulates query representations based on seasonal phase, enabling context-sensitive modeling of non-repeating fluctuations. By jointly optimizing periodic and residual pathways, PTQNet achieves accurate and interpretable forecasts with low computational overhead. Extensive experiments on eight benchmark datasets show that PTQNet outperforms eleven state-of-the-art models, with an average MSE reduction of 15.68% and an average MAE reduction of 14.88% compared to the mean of the 11 baselines. These results demonstrate that PTQNet offers a scalable and practical solution for long-horizon forecasting in resource-constrained environments.

1. Introduction

Long-term time series forecasting (LTSF) plays a pivotal role in various real-world applications, including electricity consumption prediction, inventory management, financial forecasting, and resource planning. Beyond these traditional domains, long-term forecasting over multivariate time series is increasingly central to modern information systems. Search engines, recommender systems, online platforms, and smart infrastructure generate abundant temporal logs (e.g., query volumes, click-through rates, sensor readings), whose effective processing and forecasting can reveal user and system behavior regularities, supporting core functions such as ranking, recommendation, resource allocation, and alerting. However, the inherent complexity of LTSF, such as dealing with large-scale data, long-range dependencies, and non-stationary trends, poses significant challenges to traditional forecasting methods.

Motivations. Unlike short-term forecasting that mainly leverages recent observations, long-term time series forecasting (LTSF) requires modeling extended temporal dependencies and recurrent structures that span days, weeks, or even months. For example,

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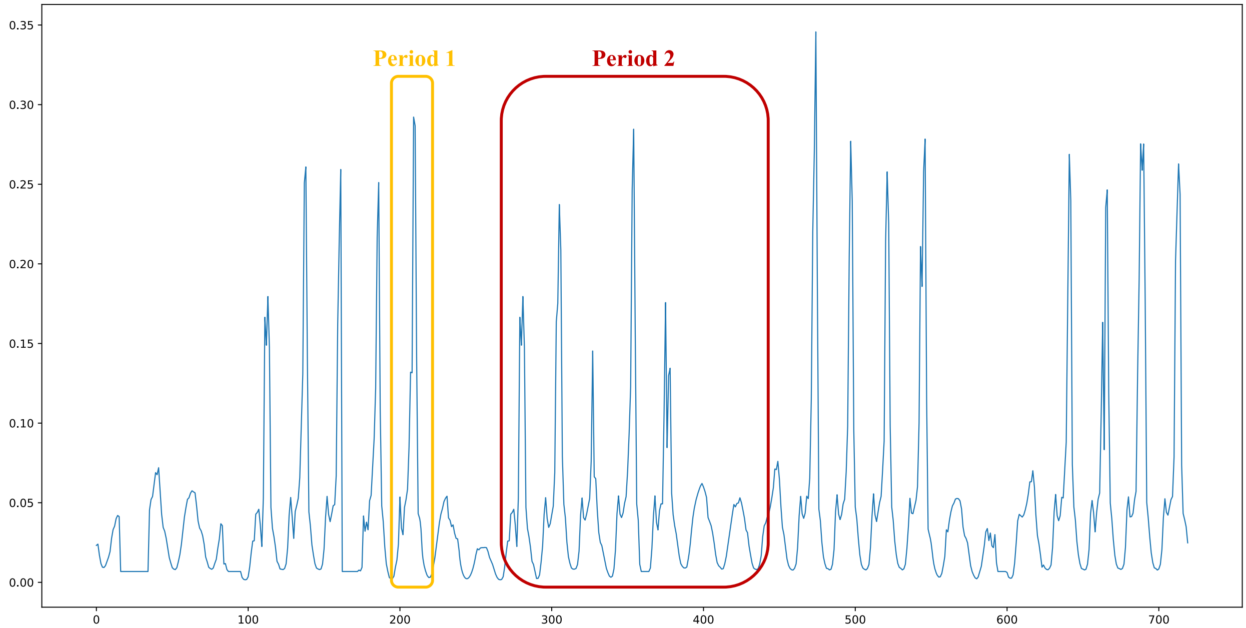

Fig. 1. Illustration of multi-scale periodicity and irregular residual fluctuations in a real-world time series segment. The deviations and spikes reflect irregular residual dynamics and anomalies that coexist with seasonality, motivating explicit periodic–residual disentanglement for long-term forecasting.

forecasting electricity demand several weeks ahead demands not only capturing local dynamics but also integrating recurring patterns such as daily (Period 1: a shorter recurring cycle) and weekly (Period 2: a longer cycle composed of repeated short cycles) cycles, as illustrated in Fig. 1. Meanwhile, real-world signals are rarely purely periodic: irregular fluctuations and anomalies (e.g., extreme weather) often coexist with structured seasonality, making it challenging to maintain both accuracy and interpretability over long horizons.

Traditional statistical approaches offer simplicity and interpretability but are often limited in their ability to capture nonlinear dynamics and cross-channel dependencies. Deep learning models, particularly recurrent architectures such as xLSTM (Beck et al., 2024), TFT (Barik & Paul, 2025), and BiGRU (Javeed et al., 2024), partially mitigate these issues but struggle with scalability in very long sequences. Recent LTSF methods span attention-based Transformer variants (e.g., Informer Zhou et al., 2021, Autoformer Wu et al., 2021, iTransformer (Liu et al., 2024), Crossformer Zhang & Yan, 2023, FEDformer Zhou et al. , 2022, PatchTST Nie et al., 2023) as well as efficient non-Transformer architectures (e.g., DLinear Zeng et al., 2023, TimeMixer Wang et al., 2024a) and multi-period/frequency-oriented designs (e.g., TimesNet Wu et al., 2023). However, even seasonality is considered, recurrent structure and irregular dynamics are often encoded within shared representations, making it difficult to expose a clear additive periodic-residual split. This makes it difficult to attribute improvements to specific cycles, and may reduce robustness when strong seasonality coexists with irregular shocks.

Typically, three representative lines of work are most relevant to our setting. TimesNet captures periodic structure through frequency-guided reshaping and representation learning, but seasonal and irregular dynamics are still encoded within shared features rather than being explicitly separated at the forecast level. CycleNet introduces learnable cyclic representations to explicitly encode periodicity, yet it does not explicitly couple seasonal phase with residual multivariate dependency modeling. TQNet improves multivariate forecasting through Temporal Query attention for inter-channel dependency learning, but it does not explicitly separate periodic and residual components before interaction modeling. These observations motivate PTQNet, which combines an explicit periodic pathway with a residual pathway whose dependency modeling is conditioned on seasonal phase.

Contributions. To address the above issues, we propose **Periodic-Temporal Query Network (PTQNet),** a lightweight framework for multivariate long-term forecasting that explicitly models multi-period structure and yields interpretable periodic representations. Specifically, PTQNet disentangles multi-scale periodic components via a hierarchical cycle memory and models irregular residual dynamics via phase-aware temporal queries, enabling accurate forecasting while retaining interpretability and efficiency. The main contributions of this work are summarized as follows:

- We propose PTQNet, which performs an explicit periodic–residual decomposition: a hierarchical cycle module learns multi-period seasonal representations from a candidate period library, while a residual pathway concentrates on irregular dynamics. This design provides an explicit periodic-residual decomposition at the pathway level and can be particularly beneficial when seasonality and irregular fluctuations coexist.

- We introduce a phase-aware residual attention mechanism that modulates temporal queries using seasonal phase information, enabling multivariate dependency modeling to adapt across phases (e.g., peak vs. off-peak) rather than assuming stationary interactions.
- We conduct extensive experiments on eight public multivariate benchmarks spanning energy systems, transportation, and finance. PTQNet delivers competitive or superior accuracy with a compact architecture, as validated by ablation and efficiency analyses, supporting its practicality on diverse long-horizon multivariate benchmarks.

2. Related work

General architectures for long-term multivariate forecasting. Recent long-term multivariate forecasting methods have adopted increasingly powerful architectures to model temporal dependencies and cross-channel interactions. Early Transformer-based models such as LogTrans (Li et al., 2020), Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021), FEDformer (Zhou et al. , 2022), and ETSformer (Woo et al., 2023) improved long-sequence modeling through sparse attention, decomposition, and classical smoothing-inspired designs. Subsequent methods further strengthened representation learning and efficiency: Pyraformer (Liu et al., 2022) and PatchTST (Nie et al., 2023) exploited structured attention or patch-based tokenization, SegRNN (Lin et al., 2023) adopted segment-wise recurrent modeling, and TiDE (Das et al., 2023), N-HiTS (Challu et al., 2022), and FITS (Xu et al., 2024) enhanced forecasting through hierarchical or multi-scale temporal modeling. More recent approaches such as STID (Shao et al., 2022), TimeMixer (Wang et al., 2024a), SparseTSF (Lin et al., 2024b), TimeXer (Wang et al., 2024c), iTransformer (Liu et al., 2024), Crossformer (Zhang & Yan, 2023), CARD (Wang et al., 2024b), and SAMformer (Ilbert et al., 2024) further improved long-horizon forecasting by combining temporal interaction, cross-dimensional dependency learning, and efficient architectural designs. These methods substantially improve forecasting performance, but recurrent seasonal structure and irregular residual dynamics are often still encoded within shared representations.

Periodicity Modeling and Decomposition. Explicit periodic modeling remains important for long-horizon forecasting. Classical decomposition-based methods such as RobustSTL (Wen et al., 2018) improve robustness to outliers, but rely on predefined decomposition assumptions. In neural forecasting, SparseTSF (Lin et al., 2024b), UnetTSF (Chu et al., 2024), UniTST (Liu et al., 2025a), and SSCNN (Deng et al., 2024) incorporate periodic kernels, downsampling, or convolutional modules to capture repetitive structures, while TimeBridge (Liu et al., 2025b), DUET (Qiu et al., 2025), DCformer (Huang et al., 2026), and PatchWaveNet (Huang et al., 2025) further emphasize decomposition, cross-scale interaction, and periodic inductive bias. TimesNet-style frequency-aware modeling captures dominant periods effectively, and CycleNet (Lin et al., 2024a) introduces learnable cyclic representations to explicitly encode periodicity. However, even in these methods, periodic structure and residual variation are typically not separated into an explicit additive periodic-residual forecasting pathway with residual modeling specialized after periodic subtraction.

Multivariate Dependency Modeling. Another closely related line of work focuses on modeling inter-variable dependencies in multivariate forecasting. Crossformer (Zhang & Yan, 2023), iTransformer (Liu et al., 2024), and related architectures strengthen channel interaction through cross-dimensional attention or variable-wise tokenization. TQNet (Lin et al., 2025) is particularly relevant to our work: it introduces a Temporal Query attention mechanism to extract global inter-channel relationships and shows robustness to noisy observations. However, its dependency modeling is performed without explicitly subtracting a periodic component beforehand. As a result, periodic structure and irregular residual dynamics remain coupled during interaction modeling.

Objectives. Motivated by the above observations, we aim to build a forecasting framework that explicitly separates multi-period seasonal structure from irregular residual dynamics while retaining effective multivariate dependency modeling. Specifically, our objectives are to: (1) model multiple recurrent cycles via learnable periodic components with adaptive weighting, avoiding overly rigid seasonal priors; (2) design a phase-aware attention mechanism so that residual dependency modeling can adapt to temporal phase/context; and (3) validate these designs across diverse datasets under a compact architectural budget.

3. Methodology

3.1. Problem definition

In multivariate time series forecasting, the problem is typically articulated as follows: Given historical observations $x_{1:\mathrm{T}} = [x_1, x_2, \dots, x_T] \in \mathbb{R}^{T\times C}$, and the objective is to predict a time series $\mathrm{x}_{\mathrm{T+1:T+L}} = [x_{T+1}, x_{T+2}, \dots, x_{T+\ell}] \in \mathbb{R}^{\ell\times C}$, where ℓ, T represent future and past time steps, respectively, and C denotes the number of features (channels). It is a natural assumption that these C sequences are interrelated (e.g., climatic characteristics of weather), and this assumption helps to improve the accuracy of the forecast.

3.2. Design rationale

The design of **PTQNet** is driven by three core issues: inflexibility in adapting to irregular periodicities due to predefined/fixed cycle lengths, insufficient mixed-seasonality modeling from weak seasonal-residual coupling, and limited scalability in high-dimensional settings from under-exploited cross-channel interactions. To this end, **PTQNet** explicitly represents multi-level periodicities through learnable recurrent cycles with adaptive weighting and gating, and integrates an adaptive convolutional layer with residual attention blocks to fuse periodic embeddings and temporal features. By dynamically adjusting to varying forecast horizons and preserving fine-grained irregular dynamics, PTQNet is designed to alleviate several limitations of prior work in adaptability to irregular periodicities, periodic-residual coordination, and computational efficiency.

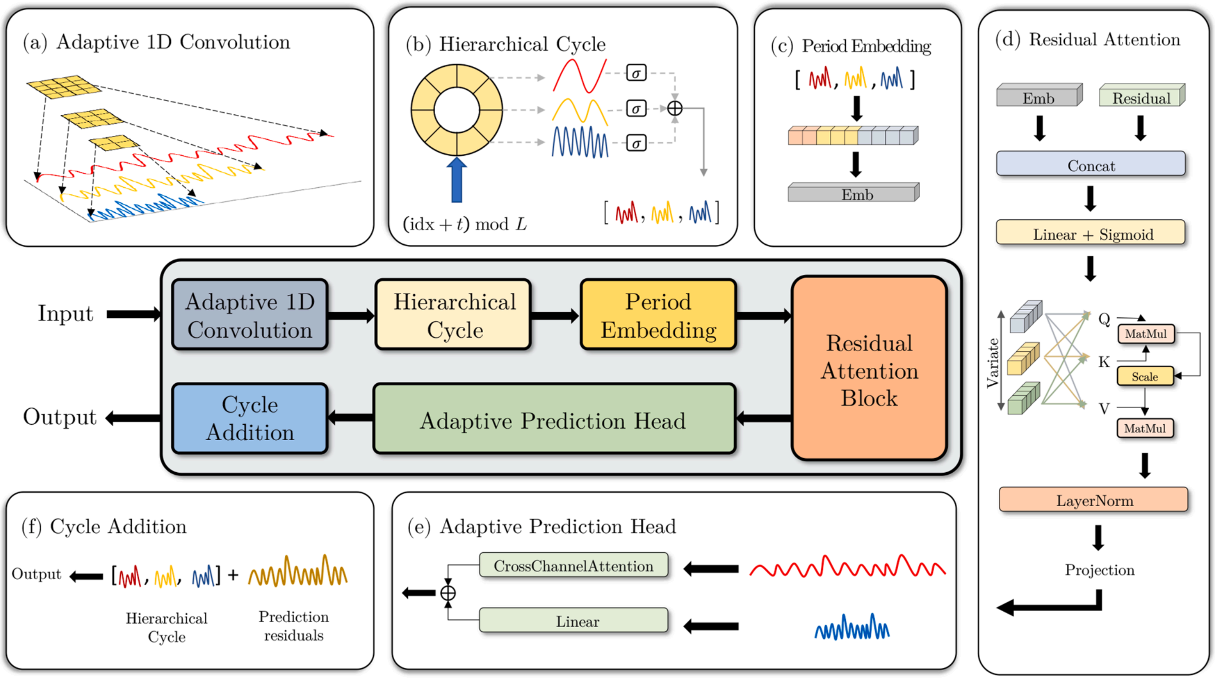

Fig. 2. Overall structure of PTQNet.

3.3. Model structure

Our proposed PTQNet is illustrated in Fig. 2. It follows an explicit decomposition–residual modeling–recomposition pipeline, comprising Adaptive 1D Convolution, Hierarchical Cycle decomposition (periodic baseline construction), phase-aware Residual Attention, an Adaptive Prediction Head, and Cycle Addition for reconstruction.

The proposed model is a hierarchical cycle-aware residual attention network tailored for long-term time series forecasting, particularly in settings characterized by multi-scale periodic structures. Given an input window, PTQNet optionally applies RevIN (Kim et al., 2021) to stabilize scale and mitigate distribution shift. The Adaptive 1D Convolution encoder then extracts short-range temporal features with a horizon-aware receptive field. Next, the Hierarchical Cycle module constructs a multi-period baseline by retrieving and aggregating cyclic embeddings from a candidate period library; this periodic component is explicitly subtracted from the encoded sequence to obtain residual signals. The residual pathway is subsequently modeled by a stack of Residual Attention blocks whose attention computation is conditioned on periodic phase/context, enabling the network to focus on irregular dynamics while remaining aligned with underlying seasonality. The Adaptive Prediction Head produces the residual forecast by fusing a linear decoder and a cross-channel attention decoder, balancing trend-like extrapolation and inter-variable dependency modeling. Finally, Cycle Addition reintroduces the periodic component (with appropriately shifted indices) to reconstruct the full forecast, followed by inverse RevIN when enabled. This design explicitly allocates periodic and residual modeling responsibilities to different pathways.

3.4. Adaptive 1D convolution

The Adaptive 1-D Convolution module plays a crucial role in enhancing the model's ability to capture local temporal dependencies before entering the deeper attention-based residual modeling. Unlike static convolutional layers commonly used in sequence models, AdaptiveConv1d is designed to adaptively adjust its kernel size based on the prediction horizon , which introduces flexibility and efficiency to the forecasting pipeline. The piecewise schedule is intended to match the effective receptive field of the convolution to the forecast horizon. For shorter horizons, a smaller kernel emphasizes recent local dynamics, whereas longer horizons benefit from a wider receptive field that aggregates information over a broader historical context. The breakpoint values used here follow the common horizon groups in long-term forecasting benchmarks and should be viewed as practical instantiations rather than universal constants. The factors such as $\frac{L}{4}$ keep the kernel size as a simple fraction of the look-back length, so that the receptive field remains within a controlled range while preserving implementation simplicity. In practice, we found that a range of nearby kernel sizes yields comparable performance, and that the main benefit arises from coupling the receptive field to the forecast horizon rather than from the exact numerical values of the schedule. The intuition is that shorter prediction horizons benefit from smaller kernels, as the model needs to focus on fine-grained recent patterns, whereas longer horizons require wider receptive fields to incorporate broader historical context. The kernel size is computed as:

$$k = \begin{cases} \min\left(\frac{L}{4}, 25\right), & \ell_{\text{pred}} \in (0, 192] \\ \min\left(\frac{L}{3}, 35\right), & \ell_{\text{pred}} \in (192, 336] \\ \min\left(\frac{L}{2}, 45\right), & \ell_{\text{pred}} > 336 \end{cases} \tag{1}$$

where L is the input sequence length, ℓ_{pred} is the prediction length. The kernel size is further adjusted to be an odd number to ensure symmetric padding and consistent output length. The convolution is applied channel-wise using:

$$\mathbf{X}_{\text{conv}} = \text{Conv1D}\left(\mathbf{X}_{\text{in}}, \text{kernel} = k, \text{padding} = \left\lfloor \frac{k}{2} \right\rfloor\right) \tag{2}$$

which ensures the output feature map has the same temporal resolution as the input, avoiding information loss from cropping or downsampling. And then, Let $X \in \mathbb{R}^{B\times C\times L}$ be the input tensor with batch size B, channel dimension C, and sequence length L. The convolution operation is:

$$\mathbf{Y}_{b,c,t} = \sum_{i=-\lfloor \frac{k}{2} \rfloor}^{\lfloor \frac{k}{2} \rfloor} w_{c,i} \cdot \mathbf{X}_{b,c,t+i} \tag{3}$$

where $w_{c,i}$ is the convolution weight for channel c at relative offset i, and $\mathbf{X}_{b,c,t+i}$ is zero-padded at boundaries if $t+i$ is out of range. The dynamic adaptation occurs through the selection of k as a function of the forecast length, making the convolution receptive field proportional to the target horizon. To maintain output dimensionality, symmetric zero-padding is applied such that the temporal resolution of the sequence is preserved post-convolution. This design allows the model to focus on short-term fluctuations for low-horizon tasks while expanding its receptive field for long-term forecasting. Empirically, this adaptation leads to improved downstream attention modeling by providing a locally-aware and length-sensitive representation of the input sequence. The AdaptiveConv1d module thus functions as a lightweight, generalizable front-end that enhances the model's expressiveness and robustness across diverse time-series scenarios.

3.5. Hierarchical cycle module with learnable period embeddings

To explicitly capture multi-scale periodic patterns inherent in time series data, a Hierarchical Cycle module has been designed and showed in Fig. 3. This module serves as a structured and learnable temporal memory. Unlike implicit periodic modeling via convolutions or attention alone, this module represents periodicities across multiple levels through direct parameterization of repeating cycles. Formally, for a set of predefined cycle lengths $\{L_1, L_2, \dots, L_k\}$,we define K separate learnable embedding tables, each of shape $L_k \times C$, where C denotes the number of input channels. These embeddings form a recurrent representation of time, where each position in a cycle learns a unique encoding reflective of its phase within that periodicity. In particular, $\{L_1, L_2, \dots, L_k\}$ denotes a small, fixed library of candidate cycle lengths measured in time steps. For each dataset, we instantiate this library using a few generic temporal scales implied by the sampling frequency, such as sub-daily, daily, and multi-day horizons (e.g., 24- and 168-step cycles and their multiples for hourly series, or the corresponding equivalents for 15-minute and higher-frequency data). This library is instantiated from a few generic temporal scales implied by the sampling frequency and kept fixed during training for each dataset, serving as an inductive bias rather than a fully data-free component.

Each cycle-level embedding is implemented via a RecurrentCycle module, which acts as a circular buffer indexed by time. For a given time index t and a sequence length ℓ, the cycle embedding for level k is retrieved by computing:

$$\mathbf{B}_k = \text{Gather}\left(\mathbf{E}_k, \left[(t+i) \ mod \ L_k\right]_{i=0}^{\ell-1}\right) \tag{4}$$

where $\mathbf{E}_k \in \mathbb{R}^{L_k\times C}$ is the learnable embedding matrix for cycle length L_k, and $\mathbf{B}_k \in \mathbb{R}^{B\times\ell\times C}$ is the resulting sequence of periodic embeddings for that level. The embeddings from different cycles are aggregated either additively or through a soft attention-weighted

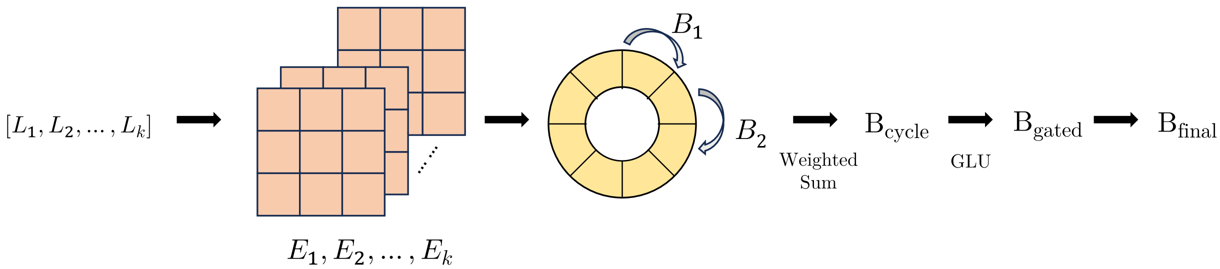

Fig. 3. Hierarchical Cycle Module. The utilization of a learnable "ring periodic table" is imperative for the explicit capture of multi-scale seasonal patterns, thereby ensuring the pristine extraction of periodic information.

sum using a learnable vector $\boldsymbol{\alpha} \in \mathbb{R}^K$:

$$\mathbf{B}_{\text{cycle}} = \sum_{k=1}^{K} \mathrm{softmax}(\boldsymbol{\alpha})_k \cdot \mathbf{B}_k \tag{5}$$

To further enhance the flexibility of this periodic representation, we introduce a GLU (Yann et al., 2017) that modulates each element of $\mathbf{B}_{\text{cycle}}$ through a learned gating mechanism. This allows the model to suppress irrelevant or noisy cyclic components while amplifying the most informative ones. The gated embedding is computed as:

$$\mathbf{B}_{\text{gated}} = \left(\mathbf{B}_{\text{cycle}} W_1 + \mathbf{b}_1\right) \odot \sigma\left(\mathbf{B}_{\text{cycle}} W_2 + \mathbf{b}_2\right) \tag{6}$$

where W_1,W_2 are learnable projection matrices, $\odot$ denotes Hadamard product and σ is the sigmoid function. This mechanism dynamically regulates the flow of periodic information based on the input state. A residual connection is optionally added to stabilize training and preserve the original periodic encoding:

$$\mathbf{B}_{\text{final}} = \mathbf{B}_{\text{cycle}} + \mathbf{B}_{\text{gated}} \tag{7}$$

The resulting output $\mathbf{B}_{\text{final}}$ serves as the refined periodic component, which is later fused with the residual prediction to form the final forecast. By incorporating GLU-based gating into the period embedding pathway, the model acquires adaptive control over the periodic signal, enhancing both expressiveness and robustness in scenarios with mixed or weak seasonality.

The advantages of this hierarchical periodic representation are threefold. Firstly, it provides an explicit inductive bias toward seasonal and repeating structures, thereby enabling the model to generalize well on datasets with known or latent periodicities. Secondly, the use of multiple concurrent cycle lengths captures complex real-world phenomena that often span multiple temporal granularities (e.g., daily, weekly, and yearly trends). Thirdly, the module offers a modular mechanism for diagnosing which candidate periodic scales are most emphasized in the learned representation by decoupling periodic encoding from other model components. Collectively, these properties designate the Hierarchical Cycle module as a malleable and methodical component for temporal modeling in long-sequence forecasting tasks.

3.6. Residual attention block

In order to effectively model irregular fluctuations that remain after periodic structure is removed, a Residual Attention Block is introduced. This block leverages self-attention to capture dynamic, non-repetitive dependencies in the residual sequence. The core design principle is to refine residual components using contextualized representations that are modulated by periodic information, thereby preserving temporal coherence while emphasizing unpredictability.

We organize a period embedding mechanism that transforms the multi-scale periodic bases learned by the hierarchical cycle module into a contextual signal for residual attention to effectively incorporate temporal structure into the residual modeling process. After subtracting the periodic components from the input sequence, the model aggregates these bases to form a fused period embedding that summarizes the dominant temporal cycles at each time step, so that the attention block focuses on residual dependency modeling after periodic subtraction. Let the hierarchical cycle module output K bases $\{B_1, B_2, \ldots, B_K\}$, each corresponding to a cycle of length L_k, where $B_K \in \mathbb{R}^{B\times\ell\times C}$ for batch size B, sequence length ℓ, and channel dimension C. When weighted aggregation is used, the model learns a set of scalar attention weights. The aggregated periodic representation is defined in Eq. 5, which will be reused in subsequent modules. To condition the attention mechanism on this periodic context, $\mathbf{B}_{\text{cycle}}$ is first projected into the attention latent space using a learnable linear transformation:

$$\mathbf{P} = \mathbf{B}_{\text{cycle}} \cdot W_p, \quad Wp \in \mathbb{R}^{C\times d} \tag{8}$$

where d is the dimension of the attention space. Simultaneously, the residual input sequence $\mathbf{X}_{\text{res}} \in \mathbb{R}^{B\times\ell\times C}$ is also projected into the same space using:

$$\mathbf{X}_{\text{proj}} = \mathbf{X}_{\text{res}} \cdot W_x, \quad W_x \in \mathbb{R}^{C\times d} \tag{9}$$

These two projected representations are then fused via a gating mechanism that produces a dynamic modulation of the residual input. This is achieved by concatenating the two and applying a sigmoid gate:

$$\mathbf{G} = \sigma\left(\left[\mathbf{X}_{\text{proj}} \parallel \mathbf{P}\right] \cdot W_g\right), \quad W_g \in \mathbb{R}^{2d\times d} \tag{10}$$

$$\mathbf{X}_{\text{mod}} = \mathbf{X}_{\text{proj}} \odot \mathbf{G} + \mathbf{P} \odot (1 - \mathbf{G}) \tag{11}$$

where $\parallel$ indicates concatenation along the last dimension. This modulation mechanism allows the model to interpolate between residual-driven and periodic-driven representations, enabling the attention module to align residual dynamics with the latent temporal rhythm.

This design introduces an explicit inductive bias toward time-aware processing, improves long-range temporal consistency, and enables better generalization across forecasting horizons. Unlike fixed positional encodings or external seasonal indicators, the period embedding is task-adaptive, data-driven, and structurally interpretable, making it a key component in PTQNet's ability to handle complex, periodic time-series data.

We denote $\mathbf{X}_{\text{mod}}$ as $\boldsymbol{Z}$, and $\mathbf{X}_{\text{proj}}$ as $\boldsymbol{Q}$. The modulated representation $\boldsymbol{Z}$ is then passed into a multi-head self-attention block to refine the residual sequence based on both local and periodic context, which enables each position in the sequence to attend to others and extract long-range dependencies. By incorporating this learned periodic embedding directly into the attention path, the model ensures that temporal regularities influence feature selection and attention weighting in a fully differentiable, end-to-end trainable manner. allowing the model to selectively amplify or suppress features based on temporal context. The output of the attention layer is combined with the original projected input via a residual connection and normalized:

$$\mathbf{H} = \text{LayerNorm}(\text{MultiHeadAttn}(\mathbf{Z}) + \mathbf{Q}) \tag{12}$$

Finally, the hidden output $\mathbf{H} \in \mathbb{R}^{B\times\ell\times d}$ is mapped back to the input channel space via a linear transformation to produce the block output and then passed to the next stage of the model.

$$\mathbf{X}^{dec} = \mathbf{H}\mathrm{W}_O + b_O, \quad W_O \in \mathbb{R}^{d\times C}, b_O \in \mathbb{R}^C \tag{13}$$

By modulating the residual input using periodic embeddings, the block ensures that attention operates with awareness of temporal structure, which is particularly valuable in long-sequence forecasting tasks where alignment with latent cycles improves accuracy. The gating function introduces a dynamic interpolation between residual and periodic signals, allowing the model to emphasize whichever source is more informative at each point in time. Moreover, the utilization of residual connections and normalization processes ensures the stability of training and facilitates the stacking of multiple such blocks without compromising performance. In essence, the Residual Attention Block facilitates the precise, context-sensitive modeling of residual dynamics. This modeling complements the hierarchical cycle representation, thereby enhancing the model's capacity to generalize across a broad spectrum of temporal patterns.

3.7. Adaptive prediction head

To flexibly accommodate different forecasting horizons and signal characteristics, we propose an Adaptive Prediction Head that fuses two complementary decoding pathways: a linear projection for extrapolating trend-like behavior and a cross-channel attention decoder for modeling complex inter-variable dependencies. This design is motivated by the observation that different forecasting tasks demand different representational emphases. For instance, autoregressive patterns or smooth local dynamics often favor direct projection, whereas non-local dependencies and variable interactions benefit from attention-based decoding. The first decoding path performs a direct linear transformation over the feature dimension:

$$\mathbf{Y}_{\text{lin}} = \mathbf{X}^{dec} W_{\text{lin}} + \mathbf{b}_{\text{lin}}, \quad W_{\text{lin}} \in \mathbb{R}^{C\times C}, \mathbf{b}_{\text{lin}} \in \mathbb{R}^C \tag{14}$$

This operation serves as a simple extrapolator that linearly transforms each timestep's latent representation into the forecast space without modeling inter-variable correlations.

The second path applies cross-channel attention to capture global dependencies across different input dimensions (channels). Each timestep $t \in \{1, \dots, \ell\}$ attends to other channels in the same timestep using a standard multi-head self-attention mechanism applied along the feature dimension. Let $\mathbf{h}_t \in \mathbb{R}^{B\times C}$ denotes the representation at time t. For each prediction step $t = 1, \dots, \ell$, let

$$U_t = \text{Embed}\left(\mathbf{X}_t^{dec}\right) \in \mathbb{R}^{B\times C\times d} \tag{15}$$

where $\text{Embed}(\cdot)$ is a learnable linear mapping that projects channel features into a d-dimensional space. We then compute the query, key, and value matrices as

$$Q_t = U_t W_Q, \qquad K_t = U_t W_K, \qquad V_t = U_t W_V,$$

with learnable parameters $W_Q, W_K, W_V \in \mathbb{R}^{d\times d}$. The channel-wise attention weights are

$$A_t = \text{Softmax}\left(\frac{Q_t K_t^\top}{\sqrt{d}}\right) \in \mathbb{R}^{B\times C\times C} \tag{16}$$

The attention output for each time step is

$$\widetilde{U}_t = A_t V_t \in \mathbb{R}^{B\times C\times d} \tag{17}$$

followed by an output projection

$$Y_{\text{attn},t} = \widetilde{U}_t W_O + b_O, \qquad W_O \in \mathbb{R}^{d\times C}, \ b_O \in \mathbb{R}^C \tag{18}$$

Finally, concatenating along the time axis gives

$$Y_{\text{attn}} = \left[\, Y_{\text{attn},1}, \dots, Y_{\text{attn},\ell} \,\right] \in \mathbb{R}^{B\times\ell\times C} \tag{19}$$

This formulation allows the model to attend to and combine feature-level signals dynamically, making it particularly effective for multivariate time series where variables influence one another in complex, non-local ways.

To combine the two forecasts, we introduce a learnable mixture coefficient $\lambda \in (0, 1)$ parameterized via a sigmoid-transformed scalar:

$$\lambda = \sigma(\omega), \quad \omega \in \mathbb{R}$$

$$\mathbf{Y} = \lambda \cdot \mathbf{Y}_{\text{attn}} + (1 - \lambda) \cdot \mathbf{Y}_{\text{lin}} \tag{20}$$

which ω is a learnable scalar parameter that controls how much weight to assign to the attention-based prediction versus the linear prediction. When ω is large and positive, meaning the model prefers the attention-based decoder. Since ω is learned during training, the model can dynamically adjust this balance depending on the dataset and task characteristics. This convex combination ensures that the output remains within a smooth interpolated space between the two decoding modes. The entire operation is differentiable and learned end-to-end during training.

This convex combination can be interpreted as interpolation between hypothesis spaces (see A for details).

Overall, the Adaptive Prediction Head serves as a robust and versatile interface between representation learning and output generation, integrating multiple modeling assumptions in a principled and trainable manner. This contributes to strong empirical performance across datasets with varying degrees of periodicity, noise, and long-range dependency.

3.8. Cycle addition

Following the isolation of the irregular residuals through periodic subtraction and subsequent processing via residual attention, the model utilizes a Cycle Addition mechanism to reinstate the original periodicity in the forecast. This reflects the assumption that periodic components are structured, learnable, and predictable, whereas the residuals capture the irregular and non-stationary behavior of the data.

The final predicted output $\mathbf{Y}_{\text{final}}$ is then reconstructed by adding the periodic signal back to the residual prediction:

$$\underbrace{\mathbf{Y}_{\text{final}}}_{\text{Full Forecast}} = \underbrace{\mathbf{B}_{\text{cycle}}}_{\text{Structured Periodic Component}} + \underbrace{\mathbf{Y}_{\text{res}}}_{\text{Irregular Component}} \tag{21}$$

Here, $\mathbf{Y}_{\text{res}}$ is the model's forecast of the non-periodic component obtained through the residual attention pathway.

From a functional perspective, the model decomposes the time series forecasting task $f(x)$ into:

$$f(x) = f_{\text{cycle}}(x) + f_{\text{res}}(x)$$

where f_{cycle} is implemented via indexable periodic memory and f_{res} is learned from data via attention-based mechanisms.

It is worth pointing out that, the additive reconstruction provides a pathway-level view of how the prediction is composed from periodic structure and learned residuals. This enables practical insights into temporal behavior. And the re-adding periodic structure ensures that the final predictions remain anchored in expected cyclical trends, preventing drift or oversmoothing in long-horizon forecasts.

In a concerted effort to enhance the precision of the final forecast and the caliber of residual modeling, a composite loss function has been adopted. This function meticulously supervises the model across two distinct levels: the complete reconstructed output and the intermediate residual component. Let $\mathbf{Y} \in \mathbb{R}^{B\times\ell\times C}$ denote the ground truth sequence. We define the training objective as:

$$\mathcal{L} = \underbrace{\text{MSE}(\mathbf{Y}_{\text{final}}, \mathbf{Y})}_{\text{Final Forecast Loss}} + \lambda_{\text{res}} \cdot \underbrace{\text{MSE}(\mathbf{Y}_{\text{res}}, \mathbf{Y} - \mathbf{B}_{\text{cycle}})}_{\text{Residual Reconstruction Loss}} \tag{22}$$

The first term measures the overall prediction error using mean squared error (MSE), encouraging the model to generate accurate forecasts after periodic addition. The second term supervises the residual pathway directly, ensuring that the residual attention blocks accurately reconstruct the non-periodic component of the signal. This decomposition-driven training strategy introduces a structural inductive bias that encourages the architecture to account explicitly for different types of temporal variation. Empirically, we find that setting $\lambda_{res} = 0.5$ provides a balanced trade-off across tasks with varying periodicity and noise.

This strategy echoes similar practices in multi-stage or decomposition-based models. For instance, Autoformer introduces a decomposition block that separates trend and seasonal components, with the loss indirectly optimized via end-to-end supervision. ETSformer builds on this by explicitly supervising exponential smoothing components during training. More generally, multi-branch architectures such as Informer and FEDformer also benefit from auxiliary or structural supervision, although they do not explicitly target residual modeling. Compared to these approaches, our method introduces a dedicated residual loss term that directly encourages accurate modeling of the irregular (non-periodic) part of the signal, resulting in a cleaner separation of modeling responsibilities.

4. Experiments

4.1. Experimental setup

All models were processed with an input length of 96 to ensure fair comparison. The batch size varies based on the dataset's scale to maximize GPU utilization while avoiding out-of-memory errors. For instance, a batch size of 16 is employed for the Traffic dataset, while 64 is utilized for the Weather dataset. The training epochs were set to 50 with early stopping based on a patience of 5 on the validation set. The dataset included prediction lengths $T \in 96, 192, 336, 720$. The number of attention heads in the multi-head attention mechanism is fixed at 4, and the dropout rate is set to 0.5 by default.

Specifically, the experiments in this paper are implemented using PyTorch (Paszke et al., 2019) and executed on a single NVIDIA GeForce RTX 4090 GPU with 32 GB memory. We utilize ADAM (Diederik et al., 2014) with an initial learning rate in $1 \times 10^{-3}, 5 \times 10^{-4}, 1 \times 10^{-4}$ and L2 loss for the model optimization. The model's embedding dimension (d_{model}) were set to 256 for the ETTh1 and ETTm2 dataset, 128 for the ETTh2 and ETTm1 dataset.

Table 1
Detailed information about the datasets.

Dataset	Timesteps	Dimensions	Cycle Patterns	Frequency	Information
ETTh1	14,400	7	Daily	Hourly	Electricity
ETTh2	14,400	7	Daily	Hourly	Electricity
ETTm1	57,600	7	Daily	15min	Electricity
ETTm2	57,600	7	Daily	15min	Electricity
Weather	52,696	21	Daily	10min	Weather
Solar	52,560	137	Daily	10min	Energy
Electricity	26,304	321	Daily & Weekly	Hourly	Electricity
Traffic	17,544	862	Daily & Weekly	Hourly	Transportation

4.2. Datasets

We evaluate the proposed method on 8 widely used real-world datasets, including the ETT series (Zhou et al., 2021), Electricity, Solar, Traffic, and Weather datasets (Wu et al., 2021). The detailed information of the datasets is shown in Table 1. These datasets span multiple application domains (energy systems, transportation, and meteorology), a range of sampling frequencies (from 10-minute Solar and Weather data to hourly ETT, Electricity, and Traffic series), and heterogeneous periodic structures (purely daily cycles versus combined daily-weekly cycles and more irregular multi-scale patterns). In addition, the dimensionality of the data varies substantially, from low-dimensional transformer temperature measurements to high-dimensional multivariate logs with hundreds of channels. This diversity enables us to assess the model's behaviour under different temporal resolutions, correlation structures, and levels of complexity that are representative of real-world decision-support scenarios. These datasets are outlined as follows.

- **ETTh1**. It is a "power transformer oil temperature" dataset with seven dimensions, collected across two years from 2016-07-01 to 2018-06-26 at a 1-h resolution.
- **ETTh2**. Similar to ETTh1, it is a "power transformer oil temperature" dataset with seven dimensions but collected from a different station at a 1-h resolution across the same time period.
- **ETTm1**. It is also based on "power transformer oil temperature" data, collected every 15 min, covering the same time range as that of ETTh1 and having similar variables.
- **ETTm2**. It is another 15-min, "power transformer oil temperature" dataset with seven dimensions, similar to ETTm1 but from a different station.
- **Electricity**. It is an "electricity consumption" dataset recording the load of 321 users, with multiple features, collected at an hourly level.
- **Traffic**. It is a traffic flow dataset with 862 variables collected at hourly resolution.
- **Weather**. It is a weather dataset containing meteorological variables such as temperature, humidity, wind speed, and precipitation, sampled at 10-minute intervals.
- **Solar**. It is a "solar power generation" dataset collected from multiple solar panels in Alabama, USA. The dataset includes meteorological and operational features such as solar irradiance, panel temperature, ambient temperature, and power output.

4.3. Baseline

Large language models (LLMs) have recently demonstrated commendable performance across diverse time-series benchmarks, yet their capabilities do not align with the objectives of this study. LLM-based forecasting approaches typically reformulate numeric sequences as tokenized sequences, leveraging extensive self-supervised pre-training to capture distributional patterns. While this strategy excels in zero-shot generalization and cross-domain transfer, it inherently couples seasonal, trend, and residual components within a single latent space, necessitating post-hoc decomposition or prompting to extract structural elements. Notably, these approaches differ fundamentally from our approach, they rely on massive external corpora and large parameter counts, and are usually deployed via inference-only or light fine-tuning. By contrast, our focus is on architectures that are trained from scratch on the target dataset under comparable parameter and compute budgets, in order to isolate the effect of architectural inductive biases. For this reason, we restrict our empirical comparison to specialized, lightweight forecasting models rather than LLM-based methods. Thus, our comparison is confined to specialized, lightweight forecasting architectures that share the same inductive bias. We select iTransformer (Liu et al., 2024), PatchTST (Nie et al., 2023), Crossformer (Zhang & Yan, 2023), FEDformer (Zhou et al. , 2022), MSGNet (Cai et al., 2023), CycleNet (Lin et al., 2024a), TimeMixer (Wang et al., 2024a), DLinear (Zeng et al., 2023), TQNet (Lin et al., 2025), TimesNet (Wu et al., 2023) and WPMixer (Mahmuddun et al., 2025) as our comparative baselines. They encompass a wide array of contemporary architectures, encompassing both decomposition-oriented models and non-decomposition competitors. Specifically, FEDformer and DLinear perform explicit trend-seasonal decomposition in the frequency and time domains, respectively. CycleNet and TimeMixer incorporate multi-scale or cycle-based representations to capture recurring patterns, and WPMixer leverages multi-resolution wavelet decomposition. The remaining baselines, including iTransformer, PatchTST, Crossformer, MSGNet, TQNet, and

TimesNet, offer compelling representatives of attention-based and MLP-style designs that do not involve an explicit periodic-residual split. Among these baselines, TimesNet, CycleNet, and TQNet are particularly important reference points because they are most closely related to PTQNet in periodic modeling, cyclic representation, and multivariate dependency modeling, respectively. This selection enables PTQNet to be evaluated against state-of-the-art decomposition strategies as well as leading non-decomposition architectures under a unified experimental protocol. All baselines are implemented using official or widely adopted open-source codebases, and hyperparameters are tuned under unified settings to ensure fair comparison. To ensure a fair comparison, we implemented and ran all baseline models under a unified experimental protocol unless otherwise noted. Hyperparameters were tuned following either the original papers or official repositories to avoid underfitting or unfair disadvantages.

4.4. Experimental evaluation

Following standard practice in long-term time series forecasting, we evaluate PTQNet using mean square error (MSE) and mean absolute error (MAE), which are the primary metrics reported by most benchmark studies. A lower MSE or MAE is indicative of superior prediction performance.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

4.5. Comparative experiment and analysis

As illustrated in Table 2, the multivariate prediction results demonstrate that PTQNet performed remarkably well on the selected datasets. In a comprehensive evaluation of 80 distinct configurations, encompassing a diverse array of datasets, evaluation metrics, and prediction lengths, PTQNet emerged as the top-performing solution in 47 instances and the runner-up in 21 cases. Overall, PTQNet achieves consistently strong performance across a wide spectrum of forecasting horizons and benchmark domains, while remaining competitive with recent state-of-the-art methods.

The enhanced performance of PTQNet is attributable to its principled architectural design, which integrates explicit periodic decomposition with a temporally contextualized attention mechanism. The model utilizes a series of inductive biases that enhance its ability to model long-term, multivariate time series in both data-efficient and computationally efficient ways.

Transformer-based models excel at capturing long-range dependencies, yet most encode seasonality, trends and irregular fluctuations in a shared latent space. This not only hinders the attribution of performance gains to specific temporal structures but also undermines model stability when seasonal patterns differ across datasets. Moreover, standard self-attention introduces non-trivial computational overhead when applied to long horizons, thus motivating architectures that allocate modelling capacity more selectively.

As previously explained, periodic components are explicitly learned and removed in PTQNet, enabling the attention mechanism to focus on residual dynamics rather than reconstructing the full signal. The residual attention block incorporates cycle-conditioned queries, thereby enabling the model to adapt its conception of correlation structure according to the seasonal phase. In contrast to the deep Transformer stacks, PTQNet employs a shallow residual attention layer, thereby facilitating faster convergence and reduced computational burden without compromising accuracy.

Linear models and MLP-based architectures demonstrate robust performance in stable, short-horizon forecasts; however, they encounter challenges in more volatile, long-horizon scenarios due to the absence of an explicit temporal structure. These models frequently depend on fixed receptive fields and static kernel sizes, which restricts their capacity to adapt to multi-scale temporal patterns. The absence of mechanisms to encode, decompose, or contextualize periodic signals renders the system susceptible to overfitting, transient noise, or misrepresentation of seasonality.

In contrast to the conventional approach of stateless feedforward models, PTQNet utilizes learnable cyclic embeddings to encode multiple recurrent structures, thereby functioning as a memory of periodic recurrence. The convolutional encoder dynamically adjusts its kernel size to match the forecast horizon, thereby enabling flexible modeling of local patterns across different timescales. The integration of periodic and residual signals is facilitated by a gating mechanism, which functions to prevent the contamination of residual learning by imperfect seasonal estimates.

Although PTQNet achieves optimal performance across most dataset in Table 2, it does not outperform all baselines in every scenario-for instance, it lags slightly behind on ETTh2 (horizon 96), ETTm2 (horizons 336/720), Traffic (horizon 192), and Weather (horizons 192/720). Notably, these performance gaps are modest, and PTQNet remains competitive across these cases. This is because that the advantages of PTQNet are most pronounced in situations where multi-scale seasonality is strong and relatively stable, and when cross-variable interactions exhibit phase-dependent structure that benefits from explicit periodic conditioning. In contrast, the above cases are likely driven by the following key factors. First, if the periodic component is weak, irregular, or frequently perturbed by exogenous variability, the marginal gain from allocating capacity to explicit periodic decomposition can be reduced, and methods emphasizing direct modeling of irregular dynamics may become comparatively favorable. Second, PTQNet relies on a predefined candidate period library. When dominant periodicities are only coarsely approximated (e.g., non-integer or slowly drifting periods), a portion of seasonal structure may be absorbed into the residual pathway, diminishing the advantage of explicit separation. Third,

Table 2
Multivariate long-term series forecasting results. The best results are highlighted in **bold** and the second best are underlined.

Dataset		ETTh1		ETTh2		ETTm1		ETTm2		Electricity		Solar		Traffic		Weather	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
iTransformer [2024]	96	0.395	0.410	0.300	0.350	0.341	0.376	0.184	0.267	0.148	0.241	0.209	0.241	0.395	0.269	0.175	0.216
	192	0.449	0.441	0.382	0.400	0.381	0.395	0.253	0.312	0.164	0.256	0.236	0.267	0.413	0.277	0.225	0.258
	336	0.492	0.465	0.424	0.432	0.419	0.419	0.315	0.352	0.178	0.270	0.248	0.274	0.422	0.282	0.280	0.298
	720	0.522	0.504	0.426	0.445	0.486	0.456	0.412	0.406	0.228	0.312	0.251	0.279	0.460	**0.302**	0.361	0.351
	Avg	0.465	0.455	0.383	0.407	0.407	0.412	0.291	0.334	0.180	0.270	0.236	0.265	0.423	**0.283**	0.260	0.281
PatchTST [2023]	96	0.380	0.399	0.298	0.347	0.328	0.367	0.183	0.268	0.180	0.273	0.234	0.287	0.459	0.284	0.176	0.217
	192	**0.425**	0.432	0.381	0.404	0.371	0.391	0.247	0.307	0.187	0.280	0.269	0.314	0.461	0.294	0.221	0.257
	336	0.470	0.458	0.422	0.438	0.398	0.409	0.314	0.350	0.214	0.308	0.294	0.318	0.483	0.321	0.279	0.297
	720	0.521	0.507	0.435	0.455	0.460	0.446	0.421	0.413	0.237	**0.276**	0.351	0.365	0.520	0.351	0.359	0.348
	Avg	0.449	0.449	0.384	0.411	0.389	0.403	0.291	0.335	0.205	0.284	0.287	0.321	0.481	0.313	0.259	0.280
Crossformer [2023]	96	0.423	0.448	0.740	0.609	0.404	0.426	0.287	0.366	0.219	0.314	0.316	0.337	0.644	0.429	0.212	0.271
	192	0.471	0.474	3.012	1.354	0.450	0.451	0.414	0.492	0.231	0.322	0.734	0.792	0.665	0.431	0.224	0.279
	336	0.570	0.575	1.973	1.105	0.603	0.559	0.597	0.542	0.246	0.337	0.758	0.746	0.674	0.420	0.284	0.351
	720	0.830	0.689	3.209	1.525	0.666	0.589	1.730	1.042	0.280	0.363	0.772	0.766	0.683	0.424	0.391	0.413
	Avg	0.574	0.547	2.234	1.148	0.531	0.506	0.757	0.611	0.244	0.334	0.645	0.660	0.667	0.426	0.278	0.329
FEDformer [2022]	96	0.395	0.424	0.358	0.397	0.379	0.419	0.203	0.287	0.193	0.308	0.267	0.351	0.587	0.366	0.221	0.305
	192	0.469	0.470	0.429	0.439	0.426	0.441	0.269	0.328	0.201	0.315	0.297	0.386	0.604	0.373	0.294	0.346
	336	0.530	0.499	0.496	0.487	0.445	0.459	0.325	0.366	0.214	0.329	0.301	0.389	0.621	0.383	0.351	0.387
	720	0.598	0.544	0.463	0.474	0.543	0.490	0.421	0.415	0.246	0.355	0.356	0.394	0.626	0.382	0.413	0.438
	Avg	0.498	0.484	0.437	0.449	0.448	0.452	0.305	0.349	0.214	0.327	0.305	0.380	0.610	0.376	0.320	0.369
MSGNet [2024]	96	0.397	0.421	0.327	0.378	0.321	0.368	0.197	0.274	0.167	0.279	0.211	0.248	0.604	0.347	0.167	0.214
	192	0.435	0.439	0.405	0.418	0.374	0.398	0.249	0.308	0.181	0.291	0.269	0.296	0.637	0.381	**0.207**	0.251
	336	0.482	0.469	0.441	0.426	0.416	0.417	0.312	0.348	0.197	0.305	0.297	0.315	0.671	0.396	0.278	0.299
	720	0.494	0.487	0.497	0.479	0.487	0.465	0.416	0.401	0.243	0.351	0.284	0.324	0.728	0.423	0.354	0.347
	Avg	0.452	0.454	0.418	0.425	0.400	0.412	0.294	0.333	0.197	0.307	0.265	0.296	0.660	0.387	0.252	0.278
CycleNet [2024]	96	0.379	0.392	**0.285**	0.335	0.326	0.364	0.166	**0.247**	0.142	0.235	0.193	0.246	0.482	0.317	0.170	0.216
	192	0.428	0.420	0.373	0.392	0.367	0.383	0.234	**0.291**	0.156	0.249	0.206	0.247	0.485	0.316	0.222	0.259
	336	0.466	**0.440**	0.421	0.433	0.396	0.401	0.294	**0.331**	0.173	**0.265**	0.219	0.268	0.486	0.308	0.276	0.297
	720	0.463	0.462	0.454	0.459	0.458	0.433	0.398	0.392	0.512	0.298	0.257	0.270	0.511	0.324	0.350	0.345
	Avg	0.434	0.429	0.383	0.405	0.387	0.395	0.273	**0.315**	0.246	0.262	0.219	0.258	0.491	0.316	0.255	0.279
TimeMixer [2024]	96	0.378	0.396	0.291	0.342	0.336	0.369	0.176	0.258	0.154	0.247	0.179	**0.223**	0.486	0.297	0.192	0.212
	192	0.431	0.425	0.368	**0.389**	0.366	0.386	0.238	0.299	0.167	0.260	0.189	0.253	0.522	0.326	0.231	0.267
	336	0.498	0.456	0.426	0.428	0.391	0.404	0.301	0.341	0.185	0.276	0.217	0.273	0.507	0.312	0.284	0.312
	720	0.497	0.477	0.451	0.458	0.459	0.446	0.393	0.397	0.226	0.312	0.220	**0.255**	0.616	0.322	0.349	0.351
	Avg	0.451	0.439	0.384	0.404	0.388	0.401	0.277	0.324	0.183	0.274	0.201	**0.251**	0.533	0.314	0.264	0.286
DLinear [2022]	96	0.397	0.412	0.340	0.394	0.346	0.374	0.213	0.293	0.210	0.302	0.296	0.374	0.650	0.396	0.197	0.261
	192	0.446	0.441	0.482	0.479	0.382	0.391	0.284	0.361	0.210	0.305	0.314	0.396	0.598	0.370	0.254	0.306
	336	0.489	0.467	0.591	0.541	0.415	0.415	0.382	0.429	0.223	0.319	0.353	0.416	0.605	0.373	0.297	0.337
	720	0.513	0.510	0.839	0.661	0.473	0.451	0.558	0.525	0.258	0.350	0.367	0.416	0.645	0.394	0.356	0.396
	Avg	0.461	0.458	0.563	0.519	0.404	0.408	0.359	0.402	0.225	0.319	0.333	0.401	0.625	0.383	0.276	0.325
TQNet [2025]	96	0.373	0.393	0.302	0.347	0.310	0.352	0.175	0.256	0.137	0.233	0.180	0.233	0.419	0.275	**0.158**	0.201
	192	0.430	0.423	0.363	0.390	0.358	0.379	0.238	0.299	0.151	**0.248**	0.190	0.247	0.432	0.276	0.208	**0.247**
	336	0.479	0.451	0.410	0.425	0.390	0.402	0.298	0.337	**0.170**	**0.265**	**0.203**	0.271	0.455	**0.279**	0.263	0.288
	720	0.510	0.489	0.433	0.446	0.455	0.441	0.399	0.396	0.213	0.297	0.212	0.267	0.497	0.314	0.346	**0.343**
	Avg	0.448	0.439	0.377	0.402	0.378	0.394	0.278	0.322	0.168	0.261	0.196	0.255	0.451	0.286	**0.244**	**0.270**
TimesNet [2023]	96	0.388	0.411	0.332	0.369	0.334	0.376	0.188	0.266	0.168	0.272	0.255	0.297	0.590	0.314	0.173	0.231
	192	0.439	0.442	0.396	0.410	0.404	0.409	0.252	0.306	0.184	0.322	0.294	0.314	0.616	0.322	0.236	0.274
	336	0.493	0.457	0.448	0.447	0.433	0.431	0.322	0.349	0.198	0.310	0.319	0.333	0.634	0.339	0.294	0.312
	720	0.516	0.493	0.434	0.448	0.484	0.460	0.418	0.405	0.220	0.320	0.337	0.339	0.659	0.349	0.362	0.373
	Avg	0.459	0.451	0.403	0.419	0.414	0.419	0.295	0.332	0.193	0.306	0.301	0.321	0.625	0.331	0.266	0.298
WPMixer [2025]	96	0.371	0.392	0.296	0.355	0.308	**0.347**	0.171	0.261	0.146	0.241	0.184	0.231	0.389	**0.268**	0.167	0.212
	192	0.429	0.432	0.364	0.399	0.365	0.383	**0.231**	0.302	0.164	0.257	0.187	0.267	**0.396**	0.281	0.216	0.254
	336	0.462	0.448	**0.366**	**0.411**	0.387	0.397	**0.276**	0.335	0.176	0.271	0.209	0.271	0.421	0.302	0.261	0.302
	720	0.459	0.471	**0.405**	0.443	0.456	0.441	**0.381**	0.399	0.237	0.314	0.234	0.283	0.466	**0.308**	**0.333**	0.352
	Avg	0.430	0.436	**0.358**	0.402	0.379	0.392	**0.265**	0.324	0.181	0.271	0.204	0.263	0.418	0.290	**0.244**	0.280

Table 2
Continued.

Dataset		ETTh1		ETTh2		ETTm1		ETTm2		Electricity		Solar		Traffic		Weather	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
PTQNet (Ours)	96	**0.361**	**0.391**	0.287	**0.330**	**0.294**	0.348	**0.163**	0.255	**0.131**	**0.232**	**0.175**	0.224	**0.387**	0.269	0.160	**0.197**
	192	0.428	**0.419**	**0.361**	0.392	**0.330**	**0.371**	**0.231**	0.294	**0.148**	0.256	**0.184**	**0.245**	0.411	**0.273**	0.219	0.250
	336	**0.438**	0.441	0.407	0.429	**0.368**	**0.389**	0.290	0.337	0.178	0.269	0.209	**0.265**	**0.413**	0.280	**0.260**	**0.287**
	720	**0.446**	**0.460**	0.421	**0.437**	**0.433**	**0.424**	0.390	**0.391**	**0.204**	0.277	**0.209**	0.269	**0.458**	0.317	0.338	0.344
	Avg	**0.418**	**0.428**	0.369	**0.397**	**0.356**	**0.383**	0.269	0.319	**0.165**	**0.259**	**0.194**	**0.251**	**0.417**	0.285	**0.244**	**0.270**

when cross-channel coupling is limited or noisy, the additional benefit of cross-channel decoding can be smaller, and per-variable temporal modeling may suffice.

In summary, the superior performance of PTQNet is attributable to its capacity to explicitly disentangle periodic structure from residual dynamics, its adaptive architecture aligned with the forecasting horizon, and its phase-aware attention mechanism that leverages both temporal position and inter-variable context. This combination enables PTQNet to generalize robustly across datasets with diverse periodic characteristics while maintaining a compact and interpretable architecture.

4.6. *Efficiency analysis*

PTQNet is designed with architectural efficiency as a central objective. Fig. 4 demonstrates the efficiency comparison between its parameter count and training speed throughput PTQNet and other mainstream models.

With regard to the training speed, PTQNet has been shown to exhibit a high level of efficiency due to its compact architecture and low arithmetic cost. On the Electricity dataset with batch size 32 and sequence length 720, PTQNet requires only 41.74 seconds per epoch. While linear and MLP-based models, such as DLinear, demonstrate slightly superior training efficiency per epoch, this advantage is accompanied by a compromise in modeling fidelity and generalization robustness.

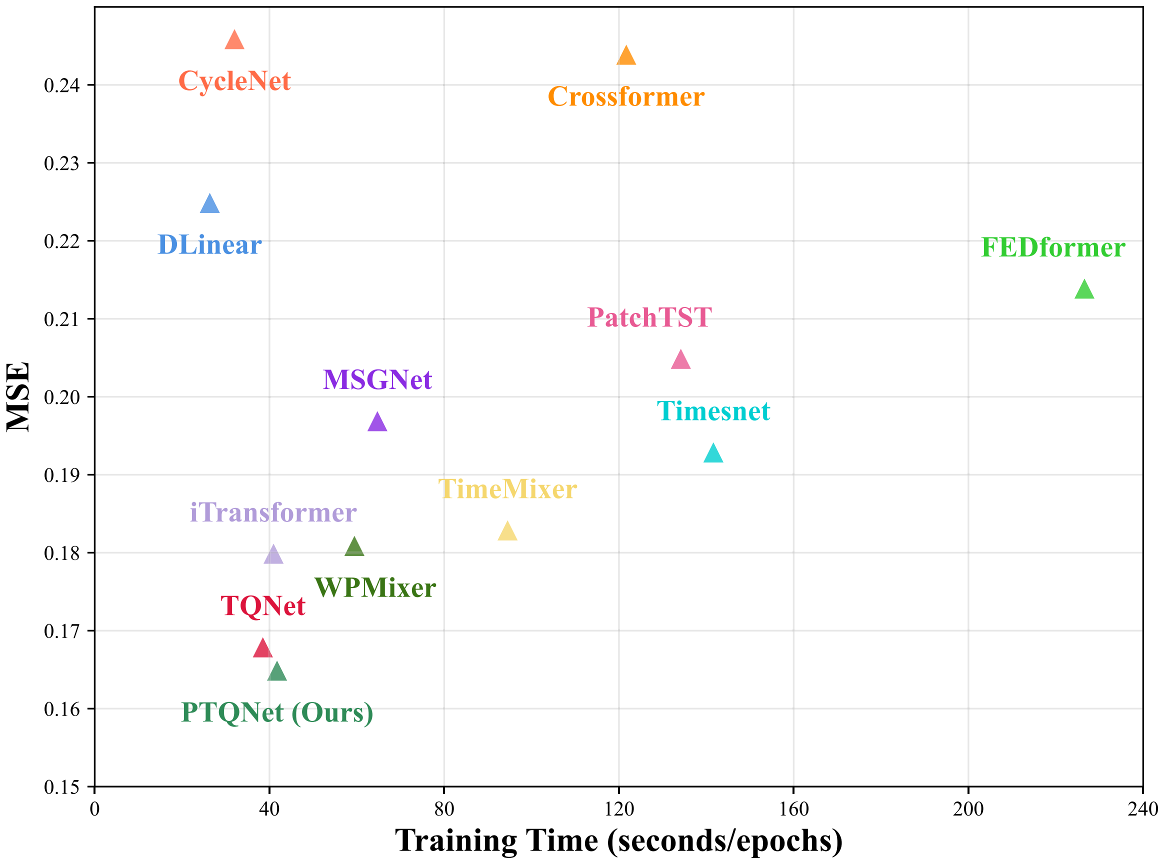

Fig. 4. Computational efficiency comparison of prediction accuracy and training time.

Table 3
Parameters and MACs comparison between PTQNet and other models on the ETTh1 dataset with look-back length $L = 96$ and forecast horizon $H = 720$..

Model	Parameters	MACs
iTransformer [2024]	4.1M	2.3G
PatchTST [2023]	6.8M	34.2G
Crossformer [2023]	5.4M	2.2G
FEDformer [2022]	16.3M	5.6G
MSGNet [2024]	4.6M	1.2G
CycleNet [2024]	1.4M	0.8G
TimeMixer [2024]	260K	0.1G
DLinear [2022]	36K	0.05G
TQNet [2025]	1.8M	0.5G
TimesNet [2023]	4.4M	1.3G
WPMixer [2025]	10.8M	6.7G
PTQNet(Ours)	1.5M	0.4G

PTQNet maintains high parameter efficiency through architectural sparsity and modular reuse. The hierarchical cycle module operates with a small number of shared embeddings, and the residual attention mechanism is deliberately shallow. As Table 3 shows, PTQNet uses approximately 1.5M parameters on the ETTh1 dataset, which is substantially lower than transformer-based models. While comparable in size to MLP-style models PTQNet incorporates stronger inductive biases for temporal structure hwithout incurring additional parameter overhead.

PTQNet avoids the quadratic scaling behavior of full self-attention by employing a single residual attention block with period-conditioned queries and adaptive convolution. On a representative configuration (sequence length = 96, prediction length = 720), PTQNet incurs 0.4G MACs, which is much more lower than most baselines, and only marginally higher than linear baselines. This reduction is attributed to the explicit disentangling of periodicity from residual learning, which mitigates the need for deep feature extraction.

PTQNet exhibits a noteworthy efficiency profile. Under the unified implementation and hardware setting used in this paper, it reduces the number of trainable parameters, lowers MACs, and shows a competitive efficiency-accuracy trade-off while preserving or exceeding accuracy. A comparison of PTQNet with MLP-based and linear baselines reveals that it maintains similar training costs while offering superior modeling capacity through its explicit decomposition of periodic and residual components. These characteristics position PTQNet as a viable and implementable solution for multivariate long-term forecasting in real world settings. We do not claim a universal speed ranking across all implementations, since absolute wall-clock throughput may vary with implementation details and hardware environments. A more comprehensive discussion of the model's computational complexity analysis can be found in B.

While PTQNet avoids the quadratic dependence on the sequence length L that characterizes deep Transformer stacks, the cross-channel attention in the prediction head introduces a quadratic term in the number of variables C. As shown in B, the overall complexity includes an $O(B\ell C^2\ d)$ component associated with modeling inter-variable interactions. In our implementation, this block is deliberately shallow and uses a moderate embedding dimension, which keeps the total computational cost manageable for the high-dimensional benchmarks considered (e.g., Electricity, Traffic, Solar). However, this design implies that PTQNet does not enjoy the strict $O(C)$ scaling of channel-independent architectures such as iTransformer or PatchTST, which treat each variable largely independently and therefore avoid cross-channel quadratic costs. The improved performance of PTQNet on multivariate benchmarks suggests that explicitly modeling cross-channel dependencies can be beneficial, but it also highlights a trade-off between expressiveness and scalability in regimes with extremely large C.

4.7. Ablation studies

To systematically evaluate the contribution of each architectural component in PTQNet, a comprehensive series of ablation studies are conducted under the setting of sequence length = 96. All experiments are conducted on the ETT datasets using identical training configurations to ensure a fair comparison. The analysis focuses on six critical aspects of the model: hierarchical cycle modeling, residual attention depth and gating, query conditioning, convolutional encoding, prediction head structure, and cycle aggregation strategy. The results are shown in Table 4.

The present study commences with an assessment of the impact of explicitly modeling multiple periodicities via the hierarchical cycle module. The first method is NoCycle, which eliminates the decomposition process and instead processes the raw sequence directly. The second method is SingleCycle, which models only one dominant period per channel. The third method is NoGate, which retains multi-period bases but removes the residual gating mechanism. The complete elimination of cycle modeling (NoCycle) results

Table 4
Ablation studies of the PTQNet. The best results are highlighted in **bold** and the second best are underlined.

Components		-		Multi-Period Decomposition with Gating						CycleConditionedQueries		Convolution		Prediction Head				CycleAggregationStrategy	
Setup		Original		No Cycle		Single Cycle		No Gate		StaticQuery		FixedConv		LinearOnly		AttnOnly		NaiveAvg	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	96	**0.371**	**0.391**	0.496	0.529	0.414	0.443	0.505	0.526	0.493	0.507	0.481	0.494	0.454	0.442	0.425	0.437	0.512	0.537
	192	**0.429**	**0.419**	0.540	0.541	0.473	0.481	0.543	0.568	0.555	0.571	0.581	0.536	0.471	0.480	0.473	0.441	0.564	0.541
	336	**0.462**	**0.441**	0.607	0.590	0.533	0.480	0.589	0.595	0.584	0.544	0.576	0.589	0.510	0.451	0.517	0.489	0.574	0.581
	720	**0.459**	**0.460**	0.609	0.612	0.534	0.495	0.586	0.613	0.582	0.593	0.601	0.612	0.545	0.542	0.491	0.494	0.573	0.599
	Avg	**0.430**	**0.428**	0.563	0.568	0.489	0.475	0.556	0.576	0.554	0.554	0.560	0.558	0.495	0.479	0.477	0.465	0.556	0.565
ETTh2	96	**0.287**	**0.330**	0.411	0.469	0.315	0.370	0.442	0.452	0.433	0.454	0.426	0.479	0.366	0.395	0.334	0.406	0.391	0.4453
	192	**0.361**	**0.392**	0.488	0.544	0.399	0.415	0.511	0.520	0.471	0.542	0.470	0.536	0.414	0.428	0.446	0.439	0.474	0.508
	336	**0.407**	**0.429**	0.514	0.567	0.467	0.465	0.555	0.549	0.518	0.547	0.538	0.548	0.441	0.502	0.456	0.485	0.561	0.535
	720	**0.421**	**0.437**	0.570	0.589	0.456	0.472	0.532	0.545	0.530	0.545	0.552	0.569	0.518	0.528	0.456	0.498	0.547	0.569
	Avg	**0.369**	**0.397**	0.496	0.542	0.401	0.439	0.510	0.517	0.488	0.522	0.497	0.533	0.435	0.463	0.423	0.457	0.493	0.514
ETTm1	96	**0.308**	**0.347**	0.460	0.459	0.375	0.394	0.426	0.453	0.456	0.484	0.431	0.462	0.381	0.422	0.366	0.426	0.445	0.500
	192	**0.365**	**0.383**	0.518	0.523	0.388	0.415	0.497	0.494	0.476	0.486	0.514	0.523	0.386	0.445	0.424	0.410	0.487	0.516
	336	**0.387**	**0.397**	0.539	0.507	0.437	0.418	0.509	0.516	0.532	0.537	0.491	0.537	0.448	0.441	0.418	0.451	0.537	0.515
	720	**0.456**	**0.441**	0.603	0.584	0.505	0.464	0.608	0.553	0.573	0.562	0.558	0.579	0.495	0.475	0.527	0.516	0.588	0.569
	Avg	**0.379**	**0.392**	0.530	0.518	0.414	0.419	0.510	0.504	0.509	0.517	0.499	0.525	0.428	0.446	0.434	0.451	0.514	0.525
ETTm2	96	**0.163**	**0.255**	0.321	0.399	0.195	0.320	0.318	0.400	0.300	0.380	0.302	0.363	0.190	0.322	0.213	0.339	0.294	0.403
	192	**0.231**	**0.294**	0.389	0.435	0.257	0.341	0.347	0.429	0.368	0.442	0.381	0.444	0.293	0.349	0.257	0.318	0.372	0.435
	336	**0.290**	**0.337**	0.399	0.479	0.339	0.391	0.403	0.485	0.440	0.490	0.415	0.471	0.355	0.403	0.337	0.388	0.427	0.460
	720	**0.390**	**0.391**	0.542	0.521	0.423	0.451	0.526	0.540	0.502	0.523	0.504	0.514	0.470	0.474	0.477	0.480	0.535	0.532
	Avg	**0.269**	**0.319**	0.413	0.459	0.305	0.369	0.399	0.464	0.403	0.459	0.401	0.448	0.327	0.387	0.321	0.381	0.407	0.458

in a substantial decline in performance. The SingleCycle model exhibits a marginal improvement but remains incapable of capturing cross-scale recurrences. Furthermore, the ablation of the gating mechanism that modulates residual strength based on each periodic basis results in increased error, thereby confirming its role in dynamic residual-path control.

In order to investigate the importance of temporal context in query construction, a comparison was made between PTQNet and a variant that uses static queries (StaticQuery). In this variant, queries are learned but do not incorporate seasonal embeddings. The degradation observed in StaticQuery suggests that period-aware query vectors facilitate better alignment between temporal structure and attention, especially in datasets with recurring seasonality. In addition, an evaluation of the contribution of the adaptive convolution module is conducted. This module is characterized by its capacity for dynamic adjustment of kernel size, a process that is based on the prediction horizon. A fixed variant (FixedConv) with a kernel size of 35 is utilized for the purpose of comparison. The findings indicate that the adaptive kernel design more effectively captures short-term variations aligned with the output horizon, thereby demonstrating enhanced local encoding in comparison to static filters.

PTQNet integrates a linear projection and a cross-channel attention mechanism to construct its prediction head. The following three variants were tested: (1) the linear branch alone (LinearOnly), (2) the cross-channel attention alone (AttnOnly). The findings indicate that both branches provide complementary information: the linear head captures trend-like global structure, while attention highlights variable interactions. Finally, a comparison is made between the default softmax-based weighted aggregation of periodic bases and naive averaging, where all cycles contribute equally. The experimental results demonstrate that weighted aggregation enables the model to adaptively emphasize more relevant frequencies, enhancing flexibility in dynamic environments.

4.8. Robustness studies

Real-world time series are frequently marred by transient fluctuations and sensor errors. In order to assess the system's robustness to such perturbations, we inject Gaussian noise with varying signal-to-noise ratios into the input sequences. As it shown in Fig. 5, in comparison with representative baselines, PTQNet exhibits a substantially diminished degree of performance degradation. This robustness is attributed to two factors. Firstly, the explicit decomposition of periodic components prevents noise amplification in the attention pathway. Secondly, the smoothing operation applied to residuals stabilizes query formation under corrupted input. However, these results should be interpreted as controlled analyses of specified perturbation or shift scenarios rather than exhaustive validation under all real-world regime shifts.

4.9. Lookback length stability

In addition to robustness across noise and irregular cycles, it is imperative to examine how forecasting accuracy evolves with different lookback lengths. A cursory examination of these phenomena reveals that brief lookback windows may prove inadequate in capturing the requisite temporal context, while their excessively protracted counterparts have the potential to introduce superfluous

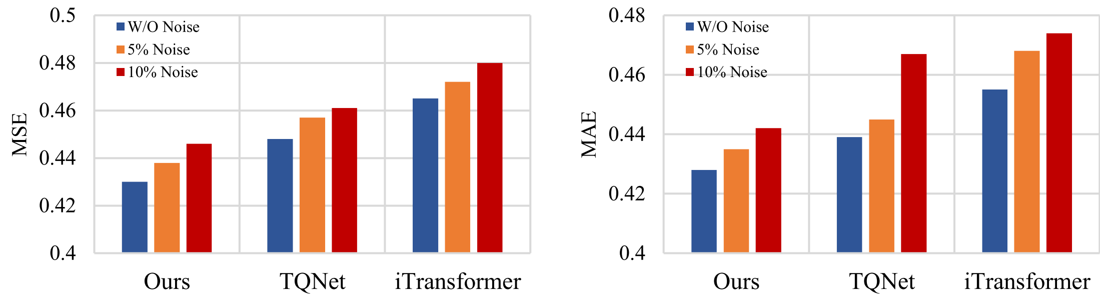

Fig. 5. Robustness analysis of different models under varying noise levels.

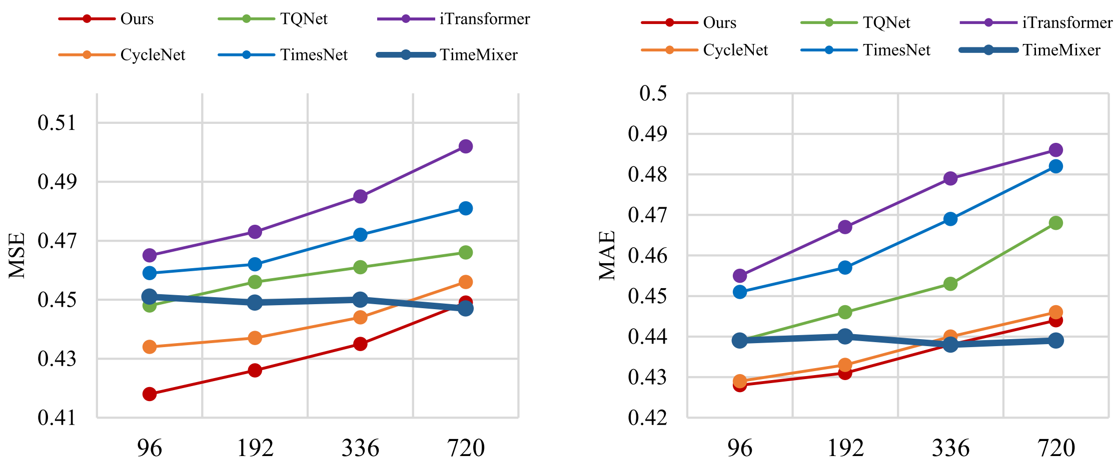

Fig. 6. Performance comparison under different lookback window lengths. Empirically, the resulting error curves indicate that PTQNet is less sensitive to increasing lookback lengths than both Transformer-based and MLP-based baselines, suggesting that explicitly separating periodic aggregation from residual modeling improves robustness to long input histories.

or disruptive information, thereby compromising the stability of the training process. To assess this, PTQNet and representative baselines are evaluated under lookback lengths 96, 192, 336, 720. Fig. 6 indicates that PTQNet exhibits consistent performance across a broad spectrum of lookback lengths, manifesting only incremental error accumulation as the context window is expanded. Conversely, deep Transformer models exhibit augmented sensitivity to lookback variation, as extended input sequences intensify error accumulation in self-attention layers. PTQNet also experiences a mild increase in error as the lookback length grows, but the slope of this degradation is substantially flatter than that of the Transformer baselines, indicating improved robustness to extended context. In contrast, the MLP-based baseline (TimeMixer) displays almost invariant performance across the tested lookback lengths, reflecting the fact that its fixed, feed-forward mixing operations are less sensitive to sequence length than deep attention stacks.

This trend stems from PTQNet's architecture, which addresses Transformer instability via two mechanisms: its hierarchical cycle module distills long-range temporal patterns into periodic embeddings, avoiding attention over extended raw sequences; its adaptive convolution limits local encoding to horizon-dependent receptive fields, curbing distant observation interference. MLP models avoid attention instability via static receptive fields but lack explicit periodic-residual separation, explaining PTQNet's superior accuracy and lower lookback sensitivity.

4.10. Sensitivity analysis

We explores the resilience of PTQNet to critical hyperparameter selections, aiming to ascertain whether the observed enhancements are attributable to sophisticated tuning or endure under reasonable configuration modifications.

Sensitivity to kernel-size schedule. We further investigate the impact of the heuristic kernel-size schedule used in the Adaptive 1D Convolution. Starting from the default piecewise rule that scales the kernel size k with the forecast horizon T and look-back length L, we consider several variants of the default kernel-size schedule. The Early-break and Late-break configurations modify the breakpoint between medium and long horizons by shifting the threshold at $T = 336$ to a smaller value ($T = 288$), so that larger kernels become active earlier, or to a larger value ($T = 384$), so that they are activated later; the short/medium breakpoint at $T = 192$ is perturbed analogously by ± 48 steps. The Small RF and Large RF variants adjust the proportional factors that map the look-back length L to kernel sizes, either shrinking the receptive field for short/medium/long horizons instead of $\frac{L}{4}, \frac{L}{3}, \frac{L}{2}$ or enlarging it, with an upper bound to ensure ($k \leq L$). Finally, the Non-adaptive variant serves as a sanity check in which the kernel size is fixed to a

Table 5
Sensitivity to the kernel-size schedule of the Adaptive 1D Convolution. We report average test MSE over horizons $T \in \{96, 192, 336, 720\}$ and the relative change $\Delta\%$ w.r.t. the default schedule.

	ETTh1		Electricity	
Schedule variant	MSE	Δ%	MSE	Δ%
Default	0.418	–	0.165	–
Early-break	0.421	+0.72%	0.167	+1.21%
Late-break	0.422	+0.96%	0.166	+0.61%
Small RF	0.424	+1.44%	0.168	+1.82%
Large RF	0.423	+1.20%	0.167	+1.21%
Non-adaptive	0.435	+4.06%	0.176	+6.67%

Table 6
Sensitivity of PTQNet to the loss-balancing coefficient λ_{res}. We report average test MSE over horizons $T \in \{96, 192, 336, 720\}$ and the relative change $\Delta\%$ w.r.t. the $\lambda_{\text{res}} = 0.5$ configuration.

	ETTh1		Electricity	
λ_{res}	MSE	Δ%	MSE	Δ%
0.0	0.457	+9.33%	0.196	+18.79%
0.2	0.436	+4.31%	0.184	+11.52%
0.4	0.428	+2.39%	0.171	+3.64%
0.5	0.418	–	0.165	–
0.6	0.425	+1.67%	0.177	+7.27%
0.8	0.441	+5.50%	0.192	+16.36%
1.0	0.460	+10.05%	0.199	+20.61%
2.0	0.516	+23.44%	0.229	+38.79%

constant fraction of L (e.g., $(k = \frac{L}{4})$) for all horizons, thereby removing horizon-dependent adaptation entirely. These configurations are designed to represent reasonable deviations from the default schedule rather than extreme, clearly suboptimal settings. As reported in Table 5, PTQNet exhibits a mild but consistent sensitivity to the choice of k, moderate deviations from the default schedule lead to small yet non-negligible increases in MSE, while the non-adaptive variant shows the largest performance degradation. These results indicate that coupling the convolutional receptive field to the forecast horizon is beneficial and that the default schedule provides a reasonable operating point. And these also suggest that what transfers beyond the standard LTSF benchmark is the horizon-aware receptive-field principle, rather than the exact breakpoint values themselves. For other applications, the same idea can be implemented by partitioning the target forecast range into short/medium/long regimes relative to the deployment horizon set and assigning progressively larger kernel sizes accordingly.

Sensitivity to the loss-balancing coefficient. The auxiliary residual loss is weighted by a scalar coefficient λ_{res}, which controls the relative emphasis on the residual pathway during training. To assess sensitivity to this hyperparameter, we evaluate PTQNet on ETTh1 and Electricity under $\lambda_{res} \in \{0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0, 2.0\}$, keeping all other settings fixed. For each configuration, we train the model with three random seeds and report the mean test MSE/MAE averaged over forecast horizons $T \in 96, 192, 336, 720$. The results in Table 6 indicate that PTQNet is not overly sensitive to the precise choice of λ_{res} within a moderate range, but exhibits a clear optimum around $\lambda_{res} = 0.5$. Settings that either suppress the residual loss entirely ($\lambda_{res} = 0$) or overweight it ($\lambda_{res} = 2.0$) lead to a mild deterioration in both MSE and MAE, whereas intermediate values ($\lambda_{res} \in [0.4, 0.6]$) yield broadly comparable performance with the best results attained at $\lambda_{res} = 0.5$. These observations suggest that the auxiliary residual objective provides a useful regularization signal when appropriately balanced against the main forecasting loss, and that PTQNet's empirical behaviour is stable around the optimal setting rather than relying on a finely tuned coefficient.

Sensitivity to periodic candidate library. To further clarify the design trade-offs of the periodic candidate library, we provide a focused sensitivity study that quantifies how library configuration affects forecasting accuracy and robustness. In particular, we examine three complementary factors: (i) the library cardinality K, (ii) the coverage range versus the candidate granularity (stride), and (iii) robustness under candidate-library misspecification where dominant periods are absent.

Period scaling across sampling rates. To ensure consistent library specification across datasets with different sampling intervals, we construct candidate periods in time units (hours) and convert them to the number of time steps. Concretely, for sampling interval Δ minutes, a real-world period of T hours corresponds to

$$P(T; \Delta) = \frac{60T}{\Delta} \text{ time steps.}$$

For example, a 24-hour cycle corresponds to 24 steps for hourly data ($\Delta = 60$), 96 steps for 15-minute data ($\Delta = 15$), and 144 steps for 10-minute data ($\Delta = 10$).

Table 7
Sensitivity to library cardinality K. We report average MSE/MAE over $\ell \in \{96, 192, 336, 720\}$. The best results are highlighted in **bold** and the second best are underlined.

Dataset	Metric	Library cardinality K					
		$K = 1$	$K = 2$	$K = 4$	$K = 6$ (Default)	$K = 8$	$K = 12$
ETTh1	MSE	0.475	0.447	0.433	**0.418**	<u>0.429</u>	0.433
	MAE	0.455	0.444	0.439	**0.428**	<u>0.438</u>	0.445
Electricity	MSE	0.195	0.177	<u>0.174</u>	**0.165**	0.176	0.178
	MAE	0.287	0.274	<u>0.266</u>	**0.259**	0.270	0.275
Traffic	MSE	0.467	0.439	0.432	**0.417**	<u>0.426</u>	0.432
	MAE	0.310	0.298	<u>0.291</u>	**0.285**	0.294	0.297
Weather	MSE	0.273	0.257	0.252	**0.244**	<u>0.250</u>	0.259
	MAE	0.296	0.283	0.279	**0.270**	<u>0.276</u>	0.283

Table 8
Sensitivity to coverage range (left) and granularity/stride (right). We report average MSE/MAE over $\ell \in \{96, 192, 336, 720\}$. The best results are highlighted in **bold** and the second best are underlined.

Dataset	Metric	Coverage range ($K = 6$)					Granularity/stride (coverage $[6, 336]$)			
		Ultra-short	Short-med	Default	Med-long	Long-only	Coarse $K = 3$	Medium $K = 6$	Fine $K = 11$	Uniform-12 $K = 12$
ETTh1	MSE	0.456	<u>0.433</u>	**0.418**	0.440	0.474	0.443	**0.418**	<u>0.427</u>	0.435
	MAE	0.450	<u>0.439</u>	**0.428**	0.444	0.460	0.445	**0.428**	<u>0.437</u>	0.440
Electricity	MSE	0.189	<u>0.175</u>	**0.165**	0.177	0.202	0.178	**0.165**	<u>0.174</u>	0.176
	MAE	0.280	<u>0.269</u>	**0.259**	0.270	0.292	0.273	**0.259**	<u>0.267</u>	0.273
Traffic	MSE	0.444	<u>0.430</u>	**0.417**	0.433	0.461	0.434	**0.417**	<u>0.425</u>	0.429
	MAE	0.301	<u>0.292</u>	**0.285**	0.295	0.310	0.296	**0.285**	<u>0.290</u>	0.296
Weather	MSE	0.265	<u>0.253</u>	**0.244**	0.257	0.276	0.261	**0.244**	<u>0.251</u>	0.256
	MAE	0.288	<u>0.277</u>	**0.270**	0.282	0.296	0.282	**0.270**	<u>0.276</u>	0.281

Experimental protocol. We conduct supplementary experiments on four representative benchmarks: ETTh1 (hourly electricity with strong diurnal patterns), Electricity (hourly consumption with daily/weekly periodicities), Traffic (hourly traffic flow with complex multi-scale seasonality), and Weather (10-minute meteorological data with diurnal cycles). We fix the input length $L = 96$ and evaluate prediction horizons $\ell \in \{96, 192, 336, 720\}$, consistent with the main experiments. We report the mean MSE/MAE averaged across the three runs.

For Library Cardinality K, we vary the number of candidate periods K while maintaining comparable temporal coverage for hourly datasets (from sub-daily to bi-weekly scales). For Weather (10-minute data), each hour value is converted to steps via $P(T; \Delta = 10)$. The following hour-based candidate sets are used: $K = 1\{24\}, K = 2 : \{24, 168\}, K = 4 : \{12, 24, 48, 168\}, K = 6$ (Default) $: \{6, 12, 24, 48, 168, 336\}, K = 8 : \{6, 12, 18, 24, 48, 96, 168, 336\}, K = 12 : \{3, 6, 9, 12, 18, 24, 36, 48, 72, 96, 168, 336\}$.

For range configurations (fixed $K = 6$), we use the following hour-based libraries: Ultra-short: $\{3, 6, 9, 12, 18, 24\}$ (sub-daily only), Short-to-medium: $\{6, 12, 24, 48, 72, 96\}$, Default: $\{6, 12, 24, 48, 168, 336\}$, Medium-to-long: $\{24, 48, 96, 168, 336, 720\}$, Long-only: $\{168, 336, 504, 720, 1008, 1440\}$ (weekly and beyond). Also, for Weather, these hour values are converted to steps using $P(T; \Delta = 10)$.

For stride configurations, we compare four candidate placements spanning the same range $[6, 336]$ hours: Coarse ($K = 3$): $\{6, 48, 336\}$; Medium ($K = 6$, Default): $\{6, 12, 24, 48, 168, 336\}$; Fine ($K = 11$): $\{6, 12, 18, 24, 36, 48, 72, 96, 144, 216, 336\}$; Uniform-12 ($K = 12$): $\{6, 36, 66, 96, 126, 156, 186, 216, 246, 276, 306, 336\}$, where Uniform-12 uses an arithmetic grid (step size 30 hours) over the same coverage.

The hourly misspecification settings are as follows:

- **R1 (ETTh1 without 24h):** remove the 24-hour candidate and keep $K = 6$ by adding a medium-scale candidate that is distinct from the existing 48h/168h anchors: $\{6, 12, 48, 72, 168, 336\}$.
- **R2 (Electricity without 24h and 168h):** remove both diurnal and weekly candidates and keep $K = 6$: $\{6, 12, 48, 72, 96, 336\}$.

We compare these against the default library $\{6, 12, 24, 48, 168, 336\}$ under the same training/evaluation protocol.

Coverage range and granularity. We next disentangle two distinct effects: (i) coverage range—whether the library includes both short and long seasonal scales—and (ii) granularity (stride)—how densely candidates are placed within a fixed range. Following Table 7, we report performance averaged over the four horizons.

Table 9
Robustness to candidate-library misspecification on real-world datasets. We report average MSE/MAE over $\ell \in \{96, 192, 336, 720\}$ (mean across three seeds). Relative changes are shown in parentheses.

Scenario	MSE		MAE	
	Default	Misspecified	Default	Misspecified
R1: ETTh1 w/o 24h	0.418	0.467 (+11.72%)	0.428	0.471 (+10.05%)
R2: Electricity w/o 24h & 168h	0.165	0.207 (+25.45%)	0.259	0.306 (+18.15%)

Robustness to library misspecification. Finally, we evaluate robustness when dominant periodicities are not present in the candidate library. We consider two practically relevant real-world settings and keep K comparable to the default to avoid confounding effects from library size.

Overall, the sensitivity results provide quantitative evidence that PTQNet's periodic candidate library is both flexible and fault-tolerant, while also clarifying where design choices matter most. First, a multi-period library is important: the minimal $K = 1$ setting consistently degrades accuracy, while performance improves rapidly as K increases and then saturates around moderate sizes ($K \in [4, 8]$), supporting the default $K = 6$ as an effective balance between parsimony and expressiveness. Second, coverage is more critical than density: omitting key short- or long-scale periodicities leads to notable degradation, whereas moderate stride variations within a well-chosen range have smaller impact; moreover, scale-aware placement is preferable to uniform sampling. Third, under misspecification where dominant periods are absent, PTQNet exhibits graceful degradation rather than complete failure, suggesting that residual modeling can alleviate though not eliminate imperfect periodic bases; however, removing multiple dominant scales reduces both accuracy and the clarity of the periodic interpretation. Based on these findings, we recommend using a compact, scale-aware library that ensures coverage of dominant domain-specific periods implied by sampling frequency, and modestly expanding coverage when periodicities are uncertain or drifting, rather than excessively densifying candidates.

4.11. Visualization of the periodic patterns learned

In order to further investigate the capacity of PTQNet to capture periodic structures of varying complexity, an analysis was conducted of the representations learned by the hierarchical cycle module across datasets with distinct temporal characteristics. These visualizations are not only diagnostic of model behaviour but also constitute an interpretability analysis of PTQNet's internal representations, which can be viewed as an interpretable information-processing layer applied to multivariate temporal data. By exposing learned cycle embeddings, horizon-dependent cycle weights, and phase-conditioned attention patterns, the hierarchical cycle module transforms high-dimensional, noisy sequences into a set of structured cyclic bases that explicitly encode dominant temporal regimes and the organisation of periodic and residual components. In doing so, the model produces compact, human-inspectable summaries of the temporal organisation of the underlying systems, which can be analysed independently of the forecasting task itself and enable analysts to connect forecast behaviour to interpretable temporal regimes. The specific periodic patterns differ, as shown in Fig. 7. This finding indicates that PTQNet does not merely memorize fixed seasonalities; rather, it learns hierarchical and adaptive representations that generalize across datasets of increasing complexity. The integration of explicit cycle embeddings with adaptive weighting enables the model to achieve two key objectives: interpretability, characterized by clear cycle decomposition, and robustness, demonstrated by multi-scale adaptivity. This validation serves to substantiate the design decision to incorporate a dedicated periodic modeling module.

The visualizations accentuate several salient properties of the proposed cycle learning mechanism:

- **Daily Pattern Learning.** In the ETTm1 dataset, the model consistently learns daily periodicity, with embeddings that exhibit distinct peak and trough structures aligned with typical electricity load fluctuations. It is noteworthy that the learned embedding captures higher activation values during daytime hours and significantly lower values during nighttime hours, reflecting human activity-driven consumption cycles. This finding indicates that the hierarchical cycle module possesses the capacity to spontaneously restore discernible and interpretable daily structures, independent of explicit supervision.
- **Multi-Scale Periodicity.** The Electricity dataset offers a more intricate context, as both daily and weekly cycles contribute to the observed dynamics. PTQNet has been demonstrated to effectively disentangle and represent these multiple temporal scales. The daily embedding captures short-term fluctuations in hourly consumption, while the weekly embedding reflects broader weekday-weekend contrasts. It is noteworthy that the learned attention weights suggest that the model assigns a weight of 0.67 to the daily cycle and 0.33 to the weekly cycle. This finding indicates that fine-grained daily variations are more predictive in this particular dataset. This dynamic weighting illustrates how PTQNet adaptively prioritizes cycles according to their predictive utility.
- **Heterogeneous Multi-Scale Learning.** The Traffic dataset represents the most challenging scenario due to its multi-scale and heterogeneous periodicity. It has been demonstrated that disparate road segments manifest markedly disparate temporal behaviors. Some sensors capture pronounced bimodal patterns, such as the morning and evening rush hours. In contrast, other sensors exhibit relatively uniform traffic flow throughout the day. PTQNet addresses this heterogeneity by learning variable-specific embeddings, where distinct sensors are associated with different cycle magnitudes and phases. Beyond the daily and weekly periodicities, the learned representations exhibit amplitude modulation effects, suggesting that the model adapts to evolving traffic intensities

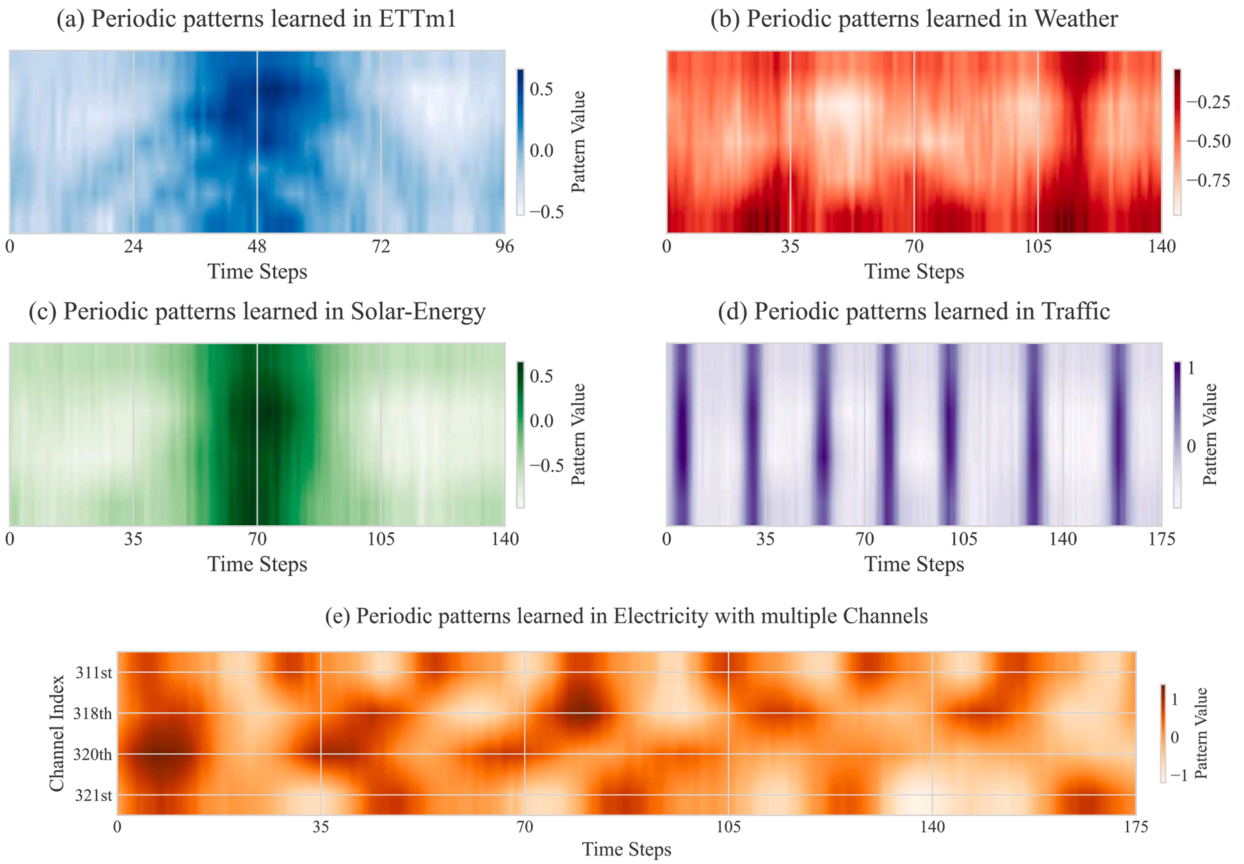

Fig. 7. Visualization of learned cyclic embeddings across datasets.

across time. This flexibility underscores PTQNet's ability to encode complex, location-dependent periodic structures within a unified architecture.

As illustrated in Fig. 8(a), the relative contributions of daily and weekly cycles undergo a transition as the prediction horizon is varied. For shorter horizons, such as 96 steps, the daily cycle exhibits a dominant influence. This finding underscores the substantial predictive capability of short-term seasonality. As the horizon extends to 720 steps, the weight assigned to the weekly cycle increases from 0.25 to 0.38, while daily cycle importance decreases to 0.53. This shift underscores the adaptive weighting mechanism employed by PTQNet, whereby longer forecasting tasks place greater reliance on broader periodic structures, while shorter tasks accentuate finer daily fluctuations.

Fig. 8(b) presents the empirical distribution of learned importance over the cycle lengths library across datasets and training runs. It is noteworthy that the log-scaled x-axis reveals that PTQNet does not collapse onto a single dominant periodicity. Rather, it maintains a diverse set of learned cycles, aligning with the multi-scale nature of real-world time series. As anticipated, the daily cycle is most frequently identified, followed by the weekly cycle. The presence of shorter cycles (12-hour and 48-hour) and longer cycles (336-hour and 720-hour) further demonstrates PTQNet's flexibility in capturing multi-resolution temporal patterns.

An analysis of the attention weights between different temporal phases is illustrated in Fig. 8(c). These phases include morning rush, business hours, evening rush, and night. The diagonal dominance indicates that PTQNet preserves strong intra-phase dependencies, while off-diagonal values highlight its ability to capture cross-phase correlations. This phenomenon assumes particular significance in the context of traffic and energy datasets, where temporally distant yet structurally related events (e.g., daily rush hours) exert influence on one another. The explicit phase-aware mechanism enhances interpretability and reinforces the architectural design choice of incorporating cycle-conditioned attention.

4.12. Visualization analysis of adaptation to irregular periodicity

A central challenge in long-term forecasting lies in the ability to adapt to non-stationary periodicities, where seasonal structures evolve over time due to external perturbations or intrinsic dynamics. To assess the resilience of PTQNet in such scenarios, we have devised a series of controlled synthetic experiments that introduce deliberate irregularities into the underlying periodic structure. Specifically, we generate multivariate sequences with abrupt shifts in cycle length and amplitude variation, thereby simulating realistic

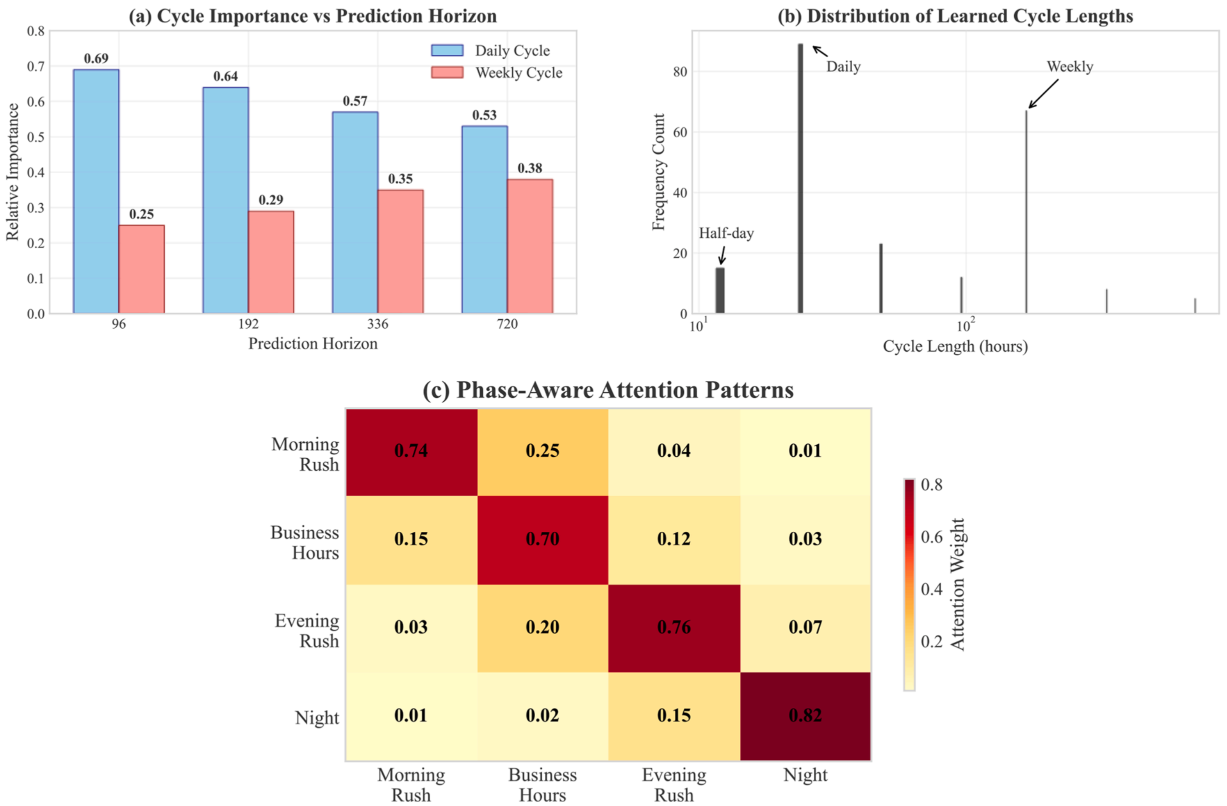

Fig. 8. Additional analysis of PTQNet's learned periodic and phase-aware behaviors. (a) Relative importance across different prediction horizons; (b) Distribution of learned importance over candidate cycle lengths across multiple training runs; (c) Phase-aware attention visualization.

non-stationary phenomena observed in domains such as energy demand under climate change or transportation patterns under policy interventions.

As demonstrated in Fig. 9(a), the synthetic series undergoes a transition from a stable 96-step cycle to a more extended 120-step cycle, with no explicit signal denoting the transition point. In response to this change, PTQNet dynamically reallocates attention among its learnable recurrent cycles (Fig. 9(b)). Specifically, the model exhibits a reduction in its reliance on the 96-cycle embedding, with the attention weight α_1 decreasing from 0.95 to 0.13, while concurrently assigning greater importance to the 120-cycle embedding, α_2 , which increases to 0.87. This adaptation unfolds smoothly over approximately 700 timesteps, thereby underscoring the model's ability to balance responsiveness with stability. In contrast to rigid decomposition-based methods, which frequently manifest brittle behavior when confronted with violated fixed seasonal assumptions, PTQNet exhibits a self-regulating adjustment mechanism that is facilitated by its hierarchical cycle module and gating operations.

The impact of this adaptation on predictive accuracy is reported in Fig. 9(c). Forecasting performance experiences a transient degradation during the transition window (timesteps 2000-2500), with Mean Absolute Error (MAE) rising from 0.345 to 0.77. Importantly, accuracy recovers to pre-shift levels (MAE $\approx$ 0.352) once the cycle reallocation process stabilizes. This recovery illustrates that PTQNet not only detects shifts in periodic structure but also learns to re-anchor its forecasting process around the new dominant cycle, thereby mitigating error accumulation across the forecast horizon.

Further insights are provided by the residual pathway analysis in Fig. 9(d). During the transition, residual magnitudes increase temporarily as the model absorbs discrepancies arising from the misalignment between old and new periodic bases. Once the periodic attention weights converge, residual magnitudes decrease, reflecting the re-establishment of coherence between the periodic embeddings and the observed sequence. This behavior validates the architectural decision to decouple periodic and residual pathways: the former provides stability and preserves structured recurrence, while the latter flexibly absorbs short-term irregularities and facilitates smooth adaptation.

Collectively, these findings confirm PTQNet's capacity to handle irregular periodicities in a stable and interpretable manner. Unlike models that implicitly encode seasonality within deep representations, PTQNet's explicit cycle-aware decomposition allows for transparent monitoring of adaptation processes and targeted resilience to evolving temporal structures. This property is especially valuable for real-world applications where periodicities are rarely stationary, ensuring that the model remains robust under structural perturbations. These shows that PTQNet's periodic and residual pathways yield interpretable intermediate representations-such as cycle profiles, regime-dependent weights, and phase-specific attention patterns-that can be examined independently of the final prediction to extract knowledge about system behaviour and its temporal organisation.

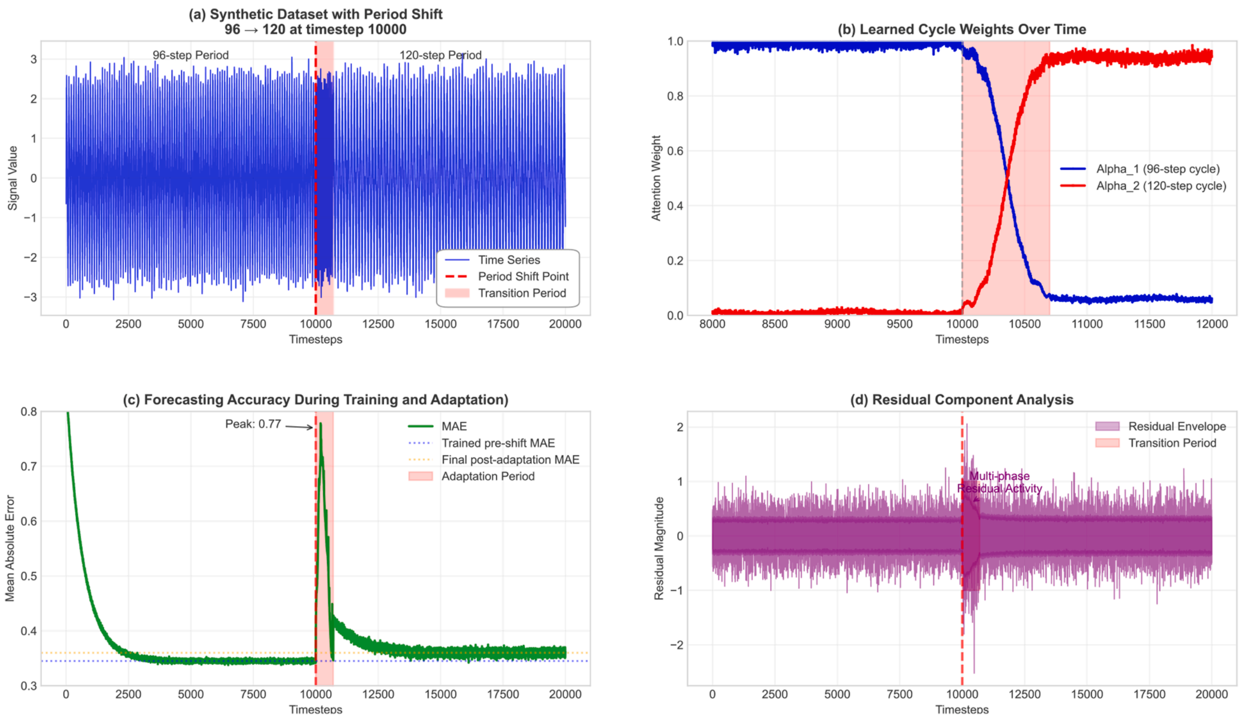

Fig. 9. PTQNet's adaptive behavior when exposed to a period shift from 96 to 120 at timestep 10,000. (a) the ground-truth synthetic dataset with an abrupt cycle transition; (b) the evolution of PTQNet's learned cycle lengths and their associated weights; (c) forecasting accuracy before, during, and after the transition; (d) residual component analysis during adaptation.

5. Discussion

The core innovation of PTQNet lies in its hierarchical decomposition strategy, which separates time series into learnable recurrent cycles and residual components. This explicit decomposition enables the model to capture structured periodicity through parameterized cyclic embeddings while dedicating its representational capacity to modeling irregular dynamics via shallow residual attention blocks. The introduction of cycle-conditioned queries further enhances this design by allowing attention mechanisms to adapt their correlation modeling based on seasonal phase, thereby achieving superior temporal coherence in long-horizon predictions.

5.1. Theoretical implications

This study contributes to the theoretical advancement of long-term time series forecasting by explicitly embedding inductive biases into model design. Unlike prior Transformer-based models that capture periodicity only implicitly through self-attention, PTQNet introduces a hierarchical cycle module that parameterizes recurrent periodicities with learnable embeddings. These components support a projection-based interpretation of periodic-residual decomposition under the assumptions discussed in C. Furthermore, the use of cycle-conditioned queries provides a theoretically grounded mechanism for aligning attention computations with seasonal phase, establishing a direct connection between temporal recurrence and correlation modeling. These innovations demonstrate that explicit periodic modeling can enhance both accuracy and robustness without the need for deep Transformer stacks. At the same time, it does not imply guaranteed recovery of true physical periodicities in noisy real-world data, nor does it constitute a full end-to-end generalization theory under practical training dynamics.

5.2. Practical applications and decision-support implications

PTQNet offers strong practical value for deployment and decision support in modern information systems. With a compact architecture (1.5M parameters, 0.4G MACs), it delivers high predictive accuracy at substantially lower computational cost, enabling use in resource-constrained settings such as smart grids, transportation monitoring, and embedded forecasting devices. The datasets used in our experiments reflect concrete decision-support scenarios that demand accurate long-horizon multivariate forecasting. For example, the Electricity and ETT datasets represent energy-system operations, where reliable forecasts of load and equipment conditions inform capacity planning, tariff design, maintenance scheduling, and demand-response strategies. In this context, PTQNet can be viewed as a general temporal processing module that transforms high-dimensional operational logs into structured forecasts for downstream decision-making, helping operators anticipate surges, detect emerging bottlenecks, and allocate resources proactively.

Beyond point forecasting, PTQNet additive decomposition into periodic and residual components and learnable multi-period embeddings. This structure enables attribution of forecast errors or anomalies to systematic cyclic variations versus irregular dynamics, clarifying whether deviations arise from shifts in recurring patterns or transient shocks. The learned periodic representations also capture daily, weekly, and seasonal regularities and their evolving importance over time, offering actionable insights into underlying system and user behavior.

5.3. Comparison with existing works

In contrast to decomposition-based methods that rely on fixed seasonal assumptions, PTQNet learns multiple cycles adaptively, allowing flexibility in handling irregular or evolving periodicities. Compared to Transformer-based approaches such as Informer, Autoformer, and PatchTST, PTQNet achieves comparable or superior accuracy with a significantly smaller architecture, due to its explicit separation of periodic and residual signals. Moreover, while existing approaches often entangle all temporal components in a shared latent space, PTQNet enforces a clear division of modeling responsibilities: periodic structure is captured via learnable embeddings, while residual dynamics are refined through attention. This principled integration distinguishes PTQNet as both more interpretable and more computationally efficient than prior work. Compared with TimesNet, which captures periodic structure through frequency-guided reshaping and representation learning, PTQNet explicitly separates periodic structure from residual dynamics at the forecasting pathway level. Compared with CycleNet, PTQNet not only models recurrent cycles explicitly, but also conditions residual multivariate dependency modeling on seasonal phase. Compared with TQNet, PTQNet performs dependency modeling after explicitly removing the periodic component, thereby allowing the residual branch to focus on non-periodic temporal variation. In this way, PTQNet combines explicit multi-period decomposition with phase-aware residual interaction modeling in a compact forecasting architecture.

5.4. Limitations and future directions

Despite its strengths, PTQNet exhibits certain limitations. First, its explicit periodic branch relies on a predefined candidate period library, so performance may degrade when dominant periods are weak, missing, non-integer, or drifting over time. Second, although PTQNet provides pathway-level interpretability, individual cycle components are not strictly identifiable in a physical sense. Third, the cross-channel attention in the prediction head scales quadratically with the number of variables and may become a bottleneck in very high-dimensional settings. In addition, it is worth noting that the current robustness evidence is based mainly on controlled perturbations and specified shift scenarios rather than exhaustive validation under all real-world regime shifts.

Future research directions include:(1) extending the hierarchical cycle module to handle adaptive or evolving periodicities through meta-learning approaches, (2) incorporating uncertainty quantification mechanisms for probabilistic forecasting, (3) investigating the application of similar decomposition principles to other temporal modeling tasks such as anomaly detection and causal inference, and (4) developing theoretical frameworks that formally characterize the expressiveness, generalization bounds, and potential identifiability conditions of explicit periodic decomposition in neural forecasting models.

5.5. Shift-aware generalization under dependence and drift

From a learning-theoretic perspective, PTQNet's generalization on high-dimensional, non-stationary time series can be understood through a shift-aware bound for dependent samples, where the future (test-window) risk is controlled by (i) an estimation/complexity term of the hypothesis class and (ii) an explicit train-test discrepancy term that captures distribution drift (and a dependence penalty under β-mixing via standard blocking arguments). Crucially, PTQNet's architecture is designed to reduce both sources of error. First, PTQNet decomposes forecasting into a low-capacity periodic module and a residual forecaster, which yields a principled complexity splitting, rather than that of a single monolithic predictor. Second, the adaptive receptive field design in the convolutional encoder directly controls the operator norm/Lipschitz constant of early feature extraction. Third, explicit periodic encoding/removal may help reduce the shift discrepancy term when the dominant seasonal structure is approximately covered by the candidate library. This aligns with our empirical period-shift experiment, where errors increase transiently during a structural change but recover after cycle reallocation stabilizes. Nevertheless, if a distribution shift introduces novel phase patterns outside the learned cycle library, missing phase support, or regime shifts not well explained by phase-induced drift, the discrepancy terms may remain large. Therefore, future improvements involve expanding the cycle library with online adaptation or meta-learned embeddings, adding explicit change-point detection to accelerate re-weighting, which should be interpreted as theoretical support for the architectural rationale rather than as a full guarantee under arbitrary training dynamics.

5.6. Broader impact

PTQNet's combination of efficiency and structured periodic-residual decomposition makes it a practical option for resource-constrained forecasting scenarios. Its compact design and pathway-level interpretability may be useful in settings where both predictive performance and a structured view of temporal variation are important. These characteristics make PTQNet particularly suitable for applications in energy management, supply chain optimization, and financial risk assessment, where both predictive accuracy and model transparency are paramount.

6. Conclusion

This paper presents PTQNet (Periodic–Temporal Query Network), a lightweight yet expressive architecture for long-term multivariate time series forecasting. This design yields a compact forecasting framework with a pathway-level periodic-residual decomposition and strong empirical performance across diverse benchmarks.

To further improve scalability and applicability, the following directions will focus on improving scalability in very high-dimensional settings through structured alternatives to the current cross-channel decoder, and on improving adaptability to evolving periodicities through more data-driven candidate-period selection and online adaptation under distribution shift.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Yaling Xun: Writing – original draft, Methodology, Data curation; **Jiahui Yan:** Validation, Investigation; **Haifeng Yang:** Writing – review & editing, Funding acquisition; **Jianghui Cai:** Writing – review & editing, Validation, Conceptualization; **Xing Wang:** Writing – review & editing.

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Appendix A. About adaptive fusion strategy

This adaptive fusion strategy provides several theoretical and practical advantages. First, it defines a forecast-dependent hypothesis class:

$$\mathcal{F}_\lambda = \lambda \cdot \mathcal{F}_{\text{attn}} + (1-\lambda) \cdot \mathcal{F}_{\text{lin}}$$

where $\mathcal{F}_{\text{attn}}$ and $\mathcal{F}_{\text{lin}}$ denote the function spaces represented by the attention and linear decoders, respectively. This formulation increases the expressiveness of the model while maintaining stability, as each component contributes a bounded and interpretable inductive bias. The learnable gating parameter λ enables task-aware decoding, effectively weighting local trend extrapolation and global dependency modeling based on the characteristics of the input data and prediction horizon. From an optimization standpoint, the presence of a linear path facilitates gradient flow and convergence, especially in early training stages where attention weights may be less stable.

Appendix B. Theoretical Complexity Analysis

Counting convention & notation. We count scalar floating point operations (FLOPs) where each scalar multiplication and each scalar addition count as one FLOP. Under this convention a matrix multiplication $\mathbb{R}^{m\times n}\times\mathbb{R}^{n\times p} \to \mathbb{R}^{m\times p}$ costs $2mnp$ FLOPs. Let B be batch size, L encoder (input) length, ℓ decoder/prediction horizon, C number of channels (variables), d the attention hidden dimension (we also write $d = hd_h$ when useful for an h-head implementation), k the convolution kernel size, and $K, \{L_k\}$ denote the number and lengths of cycle embeddings. We also report parameter counts and dominant activation-memory terms.

B.1 Adaptive 1D convolution

Depthwise (per-channel) 1D convolution with kernel length k and symmetric padding preserves the temporal length L. Each output element requires k multiplies and $(k-1)$ adds, hence $(2k-1)$ FLOPs. Therefore the exact FLOP count is

$$\text{FLOPs}_{\text{conv}} = B \cdot C \cdot L \cdot (2k-1) \approx 2\,BCLk \quad (\text{for } k \gg 1).$$

Parameters: $C \cdot k$. Activation memory (dominant): $O(BCL)$.

B.2 Hierarchical cycle module

Let $B_k \in \mathbb{R}^{B\times\ell\times C}$ be the retrieved embedding sequence for cycle k and $B_{\text{cycle}} = \sum_{k=1}^{K} \alpha_k B_k$.

Weighted aggregation. For each scalar entry the weighted sum over K cycles costs K multiplies and $(K-1)$ adds $\Rightarrow$ $(2K-1)$ FLOPs per entry, so

$$\text{FLOPs}_{\text{agg}} = B\ell C(2K-1).$$

(Computing the softmax over $\alpha \in \mathbb{R}^K$ costs $O(K)$ per position; we treat it as lower-order relative to large C or L.)

GLU refinement. We follow the paper and apply two linear projections $W_1, W_2 \in \mathbb{R}^{C\times C}$ to B_{cycle} and combine via a GLU. Each projection costs $2 \cdot B\ell \cdot C^2$ FLOPs (matrix-vector at every time-step), hence

$$\text{FLOPs}_{\text{GLU}} = 4\,B\ell C^2 + O(B\ell C)$$

(where $O(B\ell C)$ accounts for elementwise sigmoid/pointwise products).

Summary – cycle module.

$$\boxed{\text{FLOPs}_{\text{cycle}} = B\ell C(2K-1) \;+\; 4B\ell C^2 \;+\; O(B\ell C)\,.}$$

Parameters: $\sum_k L_k C \;+\; 2C^2 \;+\; 2C$ ($+K$ for α). Mem: $O(B\ell C)$.

B.3 Residual attention block (encoder-side)

We derive FLOPs for the residual attention block that operates on length L. Assumptions (explicit): input $X_{\text{res}} \in \mathbb{R}^{B\times L\times C}$; we project both X_{res} and the projected cycle embedding B_{cycle} into a shared d-dimensional attention space via $W_x, W_p \in \mathbb{R}^{C\times d}$, use a gating linear $W_g \in \mathbb{R}^{2d\times d}$, and apply standard multi-head self-attention across the temporal axis of length L (hidden dim d split into h heads). Below we count each projection and attention matmuls exactly.

Projections into attention space. Projecting X_{res} and B_{cycle}:

$$\text{FLOPs}_{\text{proj\_in}} = 2 \cdot BLCd \;+\; 2 \cdot BLCd \;=\; 4\,BLCd.$$

Gating. Applying $W_g \in \mathbb{R}^{2d\times d}$ to the concatenated vector at each time step costs

$$\text{FLOPs}_{\text{gate}} = 2 \cdot BL(2d)d = 4\,BLd^2.$$

Multi-head attention (MHA). Compute Q,K,V from Z by three linear projections ($\mathbb{R}^d \to \mathbb{R}^d$):

$$\text{FLOPs}_{\text{QKV\_proj}} = 3 \cdot 2 \cdot BLd^2 = 6\,BLd^2.$$

Attention score computation (for all heads combined) requires

$$\text{FLOPs}_{QK^\top} = 2 \cdot B \cdot L^2\, d,$$

and applying attention weights to values costs another

$$\text{FLOPs}_{AV} = 2 \cdot B \cdot L^2\, d,$$

hence for h heads with head dimension $d_{\text{head}} = d/h$:

$$\text{FLOPs}_{QK^\top} = B \times h \times L^2 \times d_{\text{head}} = BL^2\, d,$$

$$\text{FLOPs}_{AV} = B \times h \times L^2 \times d_{\text{head}} = BL^2\, d.$$

Therefore the total cost of the two attention matrix multiplications is

$$\text{FLOPs}_{\text{attn matmul}} = 2BL^2\, d.$$

Finally, output projection from concatenated heads back to d costs

$$\text{FLOPs}_{\text{attn\_out}} = 2 \cdot BLd^2.$$

Map back to channel space. Mapping the hidden d-dim output back to the channel space with $W_{\text{out}} \in \mathbb{R}^{d\times C}$ costs

$$\text{FLOPs}_{\text{map\_chan}} = 2 \cdot BLdC.$$

Block total. Summing the above terms yields (exact expression under stated assumptions)

$$\boxed{\text{FLOPs}_{\text{res\_block}} = 4\,BL^2\,d \;+\; 12\,BLd^2 \;+\; 6\,BLCd \;+\; O(BLd)\,.}$$

Explanation of grouped terms: the $4BL^2d$ term is the two attention matmuls; $12BLd^2$ collects gate and QKV/output d^2 projections; $6BLCd$ collects the input/output projections between channel and attention spaces. Parameters (block): W_x, W_p give $2Cd$, W_g gives $2d^2$, QKV and out projections add $4d^2$, and W_{out} adds dC. Total roughly

$$\#\text{params}_{\text{res}} \approx 3Cd + 6d^2.$$

Activation memory is dominated by attention scores $O(BL^2)$ and per-head intermediate tensors $O(BLd)$.

B.4 Adaptive prediction head (linear + channel-wise attention)

We adopt the lightweight per-channel implementation consistent with the paper's reported parameter budget (linear path C^2 and attention-side parameters $\approx 4Cd$). Under this design the channel-wise attention treats channels as tokens of length C and uses per-channel d-dim embeddings / projection vectors (no $d \times d$ dense QKV matrices). Below we derive FLOPs for both subpaths and the final fusion.

Linear path. Applying $W_{\text{lin}} \in \mathbb{R}^{C\times C}$ to each of ℓ time-steps costs

$$\boxed{\text{FLOPs}_{\text{lin}} = 2\,B\ell C^2.}$$

Parameters: C^2.

Channel-wise attention path (lightweight per-channel embedding). Implementation assumptions (matches paper's parameter count): for each channel c we store small vectors of length d (total parameters $\propto Cd$) used to expand the scalar channel feature into a d-dim token; Q/K/V are obtained by simple per-channel linear transforms (vectors), so parameter count for Q/K/V and final channel projection $\approx 4Cd$.

Per time-step t the dominant FLOPs are:

1. Expand/scaling (embedding): $2 \cdot BCd$ FLOPs (mult+add per entry).
2. Attention score computation $QK^{\top}$: for each batch we multiply $Q \in \mathbb{R}^{C\times d}$ by $K^{\top} \in \mathbb{R}^{d\times C}$, costing $2C^2d$ FLOPs; over the batch and ℓ steps: $2B\ell C^2 d$.
3. Applying attention weights to values AV: another $2B\ell C^2 d$.
4. Output projection from d to scalar/prediction per channel: $2B\ell Cd$.

Combine the above (grouping lower-order embedding terms into $O(B\ell Cd)$):

$$\boxed{\text{FLOPs}_{\text{chan\_attn}} = 4\,B\ell C^2\,d \;+\; O(B\ell Cd).}$$

Parameters (approx): $4Cd$ (per-channel embedding + Q/K/V + out-proj). Memory: attention score tensor per time-step is $O(BC^2)$ (per time-step), so overall $O(B\ell C^2)$ if kept; in practice scores are computed per-step or per-head and can be streamed.

Fusion. Linear and attention outputs are fused with a scalar gate λ (sigmoid), cost negligible. Therefore the head total:

$$\boxed{\text{FLOPs}_{\text{head}} = 2\,B\ell C^2 \;+\; 4\,B\ell C^2\,d \;+\; O(B\ell Cd).}$$

Parameters: $C^2 + 4Cd$ (matching the paper's stated budget). Memory: $O(B\ell d) + O(B\ell C^2)$ (attention scores/activations dominate if stored).

B.5 Cycle addition & loss

Reconstructing the final forecast $Y_{\text{final}} = B_{\text{cycle}} + Y_{\text{res}}$ is an $O(B\ell C)$ elementwise addition. The loss computation (MSE) is also $O(B\ell C)$.

B.6 Overall aggregated FLOP formula

Collecting dominant terms from the modules above and keeping leading-order terms, we obtain the model FLOP count

$$\boxed{\begin{aligned}\text{FLOPs}_{\text{PTQNet}} = {} & \underbrace{2BCLk}_{\text{adaptive conv}} + \underbrace{B\ell C(2K-1) + 4B\ell C^2}_{\text{hier. cycle}} \\ & + \underbrace{2BL^2\,d + 8BLd^2 + 4BLCd}_{\text{residual attention}} \\ & + \underbrace{4B\ell C^2\,d + 2B\ell C^2}_{\text{prediction head}} + O(B\ell C).\end{aligned}}$$

The three dominant scaling regimes are:

- If $L \gg C$ (very long encoder length) the temporal quadratic term $2BL^2d$ dominates (from the residual attention block). This aligns with the paper's design choice to keep the residual-attention stack shallow.
- If C is very large, the channel-attention cost $4B\ell C^2 d$ becomes dominant (note the $C^2 d$ scaling): inter-variable attention is the heaviest cost when dimension is high.
- For small horizons ℓ and modest C, the convolutional and cycle components (proportional to $BCLk$ and $B\ell C^2$) may dominate.

Activation memory (dominant terms).

$$\boxed{\begin{aligned}\text{Memory\_activation} \;=\;& \underbrace{O(BLC)}_{\text{input/output tensors}} \;+\; \underbrace{O(BL\cdot d)}_{\text{attention hidden states}} \\ &+\; \underbrace{O\big(B\cdot \min(L^2, L\cdot W)\big)}_{\text{attention scores, } W=\text{window size}} \\ &+\; \underbrace{O(B\ell C^2)}_{\text{channel attention scores}}\end{aligned}}$$

Appendix C. Theoretical validity and stability of the periodic-residual decomposition

PTQNet forecasts by summing (i) a periodic pathway depending only on the future time index via phase embeddings and (ii) residual pathway driven by observed history. We give function-level statements (canonical periodic component, identifiability limits), then connect them to PTQNet's parameterization.

C.1. Finite Hilbert space

Let $\{\boldsymbol{x}_t\}_{t=1}^{T}$, $\boldsymbol{x}_t \in \mathbb{R}^C$, and stack $\boldsymbol{x} := (\boldsymbol{x}_1, \dots, \boldsymbol{x}_T)$. Define the finite-dimensional Hilbert space

$$\mathcal{H}_T := \{\boldsymbol{z} = (\boldsymbol{z}_1, \dots, \boldsymbol{z}_T) : \boldsymbol{z}_t \in \mathbb{R}^C\}, \qquad \langle \boldsymbol{z}, \boldsymbol{w}\rangle_{\mathcal{H}_T} := \frac{1}{T}\sum_{t=1}^{T} \langle \boldsymbol{z}_t, \boldsymbol{w}_t\rangle_{\mathbb{R}^C},$$

with induced norm $\|\boldsymbol{z}\|^2_{\mathcal{H}_T} = \frac{1}{T}\sum_{t=1}^{T} \|\boldsymbol{z}_t\|_2^2$.

We consider the additive model used only for interpretation

$$\boldsymbol{x}_t = \boldsymbol{s}_t + \boldsymbol{u}_t + \boldsymbol{\varepsilon}_t, \qquad t = 1, \dots, T, \tag{C.1}$$

where $\boldsymbol{s}$ is periodic structure, $\boldsymbol{u}$ is irregular residual, and $\boldsymbol{\varepsilon}$ is irreducible noise.

C.2. Periodic function classes

Fix candidate periods $\mathcal{L} = \{L_1, \dots, L_K\}$ and phase maps $\phi_k(t) := t \bmod L_k$.

Definition 1 (L-periodic subspace)**.** For $L \in \mathbb{N}$,

$$\mathcal{P}(L) := \{\boldsymbol{p} \in \mathcal{H}_T : \; \boldsymbol{p}_{t+L} = \boldsymbol{p}_t \text{ whenever } t + L \le T\}.$$

Equivalently, $\boldsymbol{p} \in \mathcal{P}(L)$ iff there exists a template $\boldsymbol{e} \in (\mathbb{R}^C)^L$ such that $\boldsymbol{p}_t = \boldsymbol{e}_{\phi(t)}$ for all t, where $\phi(t) = t \bmod L$.

Define the aggregate periodic subspace

$$\mathcal{P} := \mathcal{P}(L_1) + \dots + \mathcal{P}(L_K) \subseteq \mathcal{H}_T. \tag{C.2}$$

Key periodicity fact. Let $L_\star := \mathrm{lcm}(L_1, \dots, L_K)$. Any $\boldsymbol{p} \in \mathcal{P}$ is $L_\star$-periodic on $\{1, \dots, T\}$, hence $\mathcal{P} \subseteq \mathcal{P}(L_\star)$. (This is immediate since each $\mathcal{P}(L_k) \subseteq \mathcal{P}(L_\star)$.)

C.3. Canonical periodic–residual decomposition and when it matches "truth"

Canonical decomposition. Regardless of any generative assumption, the canonical periodic component induced by $\mathcal{P}$ is the orthogonal projection $\Pi_{\mathcal{P}}\boldsymbol{x}$.

Theorem 1 (Projection is the unique best periodic approximation)**.** *For any $\boldsymbol{x} \in \mathcal{H}_T$, $\boldsymbol{s}^\dagger := \Pi_{\mathcal{P}}\boldsymbol{x}$ is the unique minimizer of*

$$\min_{p\in\mathcal{P}} \|\boldsymbol{x} - \boldsymbol{p}\|_{\mathcal{H}_T}.$$

Moreover, for any $\boldsymbol{s} \in \mathcal{P}$,

$$\|\boldsymbol{x} - \boldsymbol{s}\|^2_{\mathcal{H}_T} = \|\boldsymbol{x} - \boldsymbol{s}^\dagger\|^2_{\mathcal{H}_T} + \|\boldsymbol{s} - \boldsymbol{s}^\dagger\|^2_{\mathcal{H}_T}.$$

Theorem 2. *Orthogonal projection onto a finite-dimensional subspace exists and is unique. Let $\boldsymbol{s}^\dagger = \Pi_{\mathcal{P}}\boldsymbol{x}$ and $\boldsymbol{r} := \boldsymbol{x} - \boldsymbol{s}^\dagger \in \mathcal{P}^\perp$. For any $\boldsymbol{s} \in \mathcal{P}$, write $\boldsymbol{x} - \boldsymbol{s} = \boldsymbol{r} + (\boldsymbol{s}^\dagger - \boldsymbol{s})$ with orthogonal summands $\boldsymbol{r} \in \mathcal{P}^\perp$ and $(\boldsymbol{s}^\dagger - \boldsymbol{s}) \in \mathcal{P}$, giving the Pythagorean identity and optimality.*

When does the canonical component equal the ground-truth periodic signal. If the data admit a decomposition whose residual has no component in $\mathcal{P}$, then the canonical periodic component recovers the true one.

Assumption 1 (Well-specified periodicity and orthogonal residual)**.** There exists $\boldsymbol{x} = \boldsymbol{s}^\star + \boldsymbol{u}^\star$ with $\boldsymbol{s}^\star \in \mathcal{P}$ and $\boldsymbol{u}^\star \in \mathcal{P}^\perp$.

Corollary 1 (Uniqueness under orthogonality)**.** *Under Assumption 1, $\boldsymbol{s}^\star = \Pi_{\mathcal{P}}\boldsymbol{x}$ and $\boldsymbol{u}^\star = \boldsymbol{x} - \Pi_{\mathcal{P}}\boldsymbol{x}$, hence the decomposition is unique.*

Corollary 2. *By Theorem 1, $\Pi_{\mathcal{P}}\boldsymbol{x} \in \mathcal{P}$ and $\boldsymbol{x} - \Pi_{\mathcal{P}}\boldsymbol{x} \in \mathcal{P}^\perp$. If also $\boldsymbol{x} = \boldsymbol{s}^\star + \boldsymbol{u}^\star$ with $\boldsymbol{s}^\star \in \mathcal{P}$ and $\boldsymbol{u}^\star \in \mathcal{P}^\perp$, then $\boldsymbol{s}^\star - \Pi_{\mathcal{P}}\boldsymbol{x} = \boldsymbol{u} - \boldsymbol{u}^\star$ lies in $\mathcal{P} \cap \mathcal{P}^\perp = \{\boldsymbol{0}\}$.*

Misspecification boundary. If Assumption 1 fails (missing periods or phase-locked residual), then $\boldsymbol{s}^\dagger = \Pi_{\mathcal{P}}\boldsymbol{x}$ remains the best periodic approximation in $\mathcal{P}$, and the irreducible mismatch is $\|\boldsymbol{x} - \boldsymbol{s}^\dagger\|_{\mathcal{H}_T}$.

C.4. Connection to PTQNet periodic pathway and what it can represent.

PTQNet uses embeddings $\boldsymbol{E}_k \in \mathbb{R}^{L_k \times C}$ with lookup $\boldsymbol{b}_k(t) = \boldsymbol{E}_k[\phi_k(t)]$ and mixture weights $\boldsymbol{\pi} = \mathrm{softmax}(\boldsymbol{\alpha})$:

$$\boldsymbol{b}(t) = \sum_{k=1}^{K} \pi_k \boldsymbol{b}_k(t).$$

Lemma 1 (Embedding lookup parameterizes $\mathcal{P}(L)$)**.** *Fix L. The map $\boldsymbol{E} \mapsto \boldsymbol{p}$ with $\boldsymbol{p}_t = \boldsymbol{E}[\phi(t)]$ produces exactly $\mathcal{P}(L)$.*

Lemma 2. *If $\boldsymbol{p}_t = \boldsymbol{E}[\phi(t)]$, then $\phi(t+L) = \phi(t)$ so $\boldsymbol{p}_{t+L} = \boldsymbol{p}_t$. Conversely, for any $\boldsymbol{p} \in \mathcal{P}(L)$, define $\boldsymbol{E}[j]$ as the common value of $\boldsymbol{p}_t$ over times with $\phi(t) = j$; periodicity ensures this is well-defined.*

Thus the pre-gating periodic branch lies in $\mathcal{P}$ (as a function over t). PTQNet then applies a pointwise gate $g(t) = f(\boldsymbol{b}(t))$ (channel-wise affine + sigmoid product), and outputs $\boldsymbol{b}_{\mathrm{final}}(t) = \boldsymbol{b}(t) + g(t)$. This nonlinear step generally leaves the linear subspace $\mathcal{P}$, but it preserves periodicity:

Proposition 1 (Gating preserves periodicity)**.** *Let $L_\star = \mathrm{lcm}(L_1, \dots, L_K)$. Then $\boldsymbol{b}(\cdot)$ is $L_\star$-periodic, and for any deterministic pointwise map f, $g(t) = f(\boldsymbol{b}(t))$ and $\boldsymbol{b}_{\mathrm{final}}(t) = \boldsymbol{b}(t) + g(t)$ are also $L_\star$-periodic. Hence the PTQNet periodic output belongs to $\mathcal{P}(L_\star)$ (not necessarily to $\mathcal{P}$).*

Proposition 2. *Each $\boldsymbol{b}_k(\cdot)$ is L_k-periodic, hence $L_\star$-periodic. Therefore their convex combination $\boldsymbol{b}(\cdot)$ is $L_\star$-periodic. If $\boldsymbol{b}(t+L_\star) = \boldsymbol{b}(t)$, then $g(t+L_\star) = f(\boldsymbol{b}(t+L_\star)) = f(\boldsymbol{b}(t)) = g(t)$, and the same holds for the sum.*

C.5. Inference-time stability

A central stability benefit is architectural: the periodic branch depends only on time indices and trained parameters.

Proposition 3 (Zero sensitivity to input-value perturbations at inference)**.** *Fix trained periodic-branch parameters ($\{\boldsymbol{E}_k\}, \boldsymbol{\alpha}$, gate weights). For any two histories $\boldsymbol{x}_{1:T}$ and $\tilde{\boldsymbol{x}}_{1:T}$ (arbitrarily different), the periodic-branch output at any time index t is identical:*

$$\boldsymbol{b}_{\mathrm{final}}(t; \boldsymbol{x}_{1:T}) = \boldsymbol{b}_{\mathrm{final}}(t; \tilde{\boldsymbol{x}}_{1:T}).$$

Equivalently, the mapping from observed values to periodic output has Lipschitz constant 0.

Proposition 4. *$\boldsymbol{b}_{\mathrm{final}}(t)$ is computed from $\{\phi_k(t)\}$ and fixed parameters only; it does not use $\boldsymbol{x}_{1:T}$.*

C.6. Overlapping periods

Total periodic signal vs. per-period components. The canonical total periodic component $\Pi_{\mathcal{P}}\boldsymbol{x}$ is unique (Theorem 1). However, decomposing a given $\boldsymbol{s} \in \mathcal{P}$ into $\boldsymbol{s} = \sum_k \boldsymbol{s}^{(k)}$ with $\boldsymbol{s}^{(k)} \in \mathcal{P}(L_k)$ is generally not unique.

Lemma 3 (Intersections are unavoidable)**.** *For any L_1, L_2, constant sequences belong to $\mathcal{P}(L_1) \cap \mathcal{P}(L_2)$; hence per-period splitting is non-identifiable without additional constraints whenever $K \geq 2$. More generally, on bi-infinite sequences (or when T spans at least one $\mathrm{lcm}(L_1, L_2)$ block),*

$$\mathcal{P}(L_1) \cap \mathcal{P}(L_2) = \mathcal{P}(\gcd(L_1, L_2)).$$

Lemma 4. *If $\boldsymbol{p}_t \equiv \boldsymbol{c}$, then $\boldsymbol{p}_{t+L} = \boldsymbol{p}_t$ for all L, so $\boldsymbol{p} \in \mathcal{P}(L_1) \cap \mathcal{P}(L_2)$. The gcd claim is standard: a sequence that is both L_1- and L_2-periodic is $\gcd(L_1, L_2)$-periodic.*

Canonical per-period split via a strictly convex criterion. Even when identifiability fails, one can define a unique split by selecting the minimum-energy allocation:

$$(s^{(1)},\dots,s^{(K)}) := \arg\min_{p^{(k)}\in\mathcal{P}(L_k)} \sum_{k=1}^{K} \|p^{(k)}\|_{\mathcal{H}_T}^2 \quad \text{s.t.} \quad \sum_{k=1}^{K} p^{(k)} = s. \tag{C.3}$$

Theorem 3 (Uniqueness of minimum-energy split)**.** *For any fixed $s\in\mathcal{P}$, (C.3) has a unique solution.*

Theorem 4. *The feasible set is nonempty and affine in the finite-dimensional product space $\prod_k \mathcal{P}(L_k)$. The objective is strictly convex (sum of squared norms), hence admits a unique minimizer over any nonempty affine set.*

Interpretation: ℓ_2 regularization (e.g., weight decay on embeddings) induces a stable, canonical allocation even under heavy overlap.

C.7. Boundary behavior

In the single-period template setting above, if two indices t,t' share the same phase $\phi(t)=\phi(t')=j$, then $\hat{s}_{\phi(t)}=\hat{s}_j=\hat{s}_{\phi(t')}$ by construction. Thus the estimation uncertainty of the periodic component depends on phase counts, not forecast distance.

Assumption 2 (Noise and residual conditions for concentration)**.** Assume $\{\varepsilon_t\}_{t=1}^T$ are independent, mean-zero, σ-sub-Gaussian:

$$\mathbb{E}\left[e^{\lambda\varepsilon_t}\right] \le \exp(\sigma^2\lambda^2/2) \quad \text{for all } \lambda\in\mathbb{R}, \text{ and all } t.$$

Assume the residual satisfies $\mathbb{E}[u_t \mid \phi(t)=j]=0$ for each phase j (i.e., the residual has no phase-locked mean component).

Proposition 5 (Phase variance is horizon-invariant)**.** *Under Assumption 2,*

$$\mathrm{Var}(\hat{s}_{\phi(t)} \mid \{\phi(\tau)\}_{\tau=1}^T) = \mathrm{Var}(\hat{s}_j \mid \{\phi(\tau)\}_{\tau=1}^T) \le \frac{\sigma^2+\tau^2}{n_j},$$

for any past or future t with $\phi(t)=j$.

Proposition 6. *$\hat{s}_{\phi(t)}=\hat{s}_j$ whenever $\phi(t)=j$. The bound follows from the variance of an average of mean-zero independent sub-Gaussian variables (dominated by $(\sigma^2+\tau^2)/n_j$; exact for Gaussian).*

C.8. Summary

- **Canonical periodic component is always unique.** $\Pi_{\mathcal{P}}x$ is the unique best approximation in $\mathcal{P}$. It equals the ground-truth periodic signal only under the standard orthogonality condition.
- **Per-period components are generally not identifiable.** overlaps are unavoidable. A unique, stable split can be defined via a strictly convex minimum-energy criterion, consistent with ℓ_2 regularization.
- **Stability.** periodic-branch inference output is exactly noise-insensitive. Template learning concentrates at rate $O(1/\sqrt{n_{\min}})$ under explicit concentration assumptions. Ridge estimation is uniquely defined and is a strict contraction in x.
- **Boundary behavior.** periodic uncertainty depends on phase counts, not horizon distance.

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